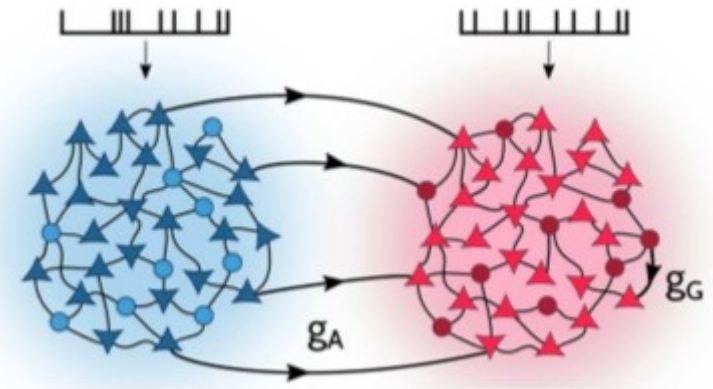
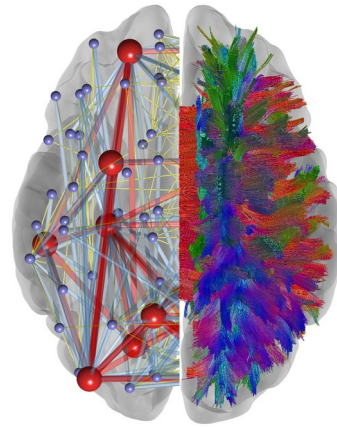
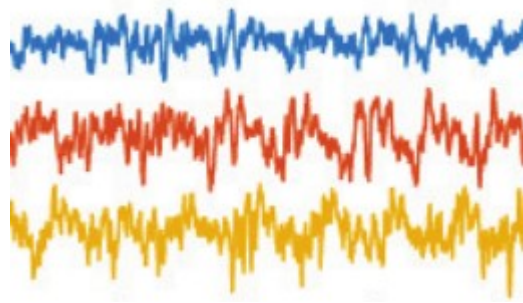


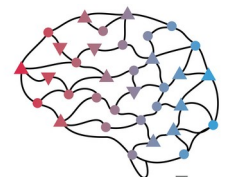


Phase relation diversity in neuronal populations

Fernanda Selingardi Matias

Instituto de Física, Universidade Federal de Alagoas, Maceió, Brasil

fernanda@fis.ufal.com



NeuroLab
[@neurolab.if.ufal](https://twitter.com/neurolab.if.ufal)

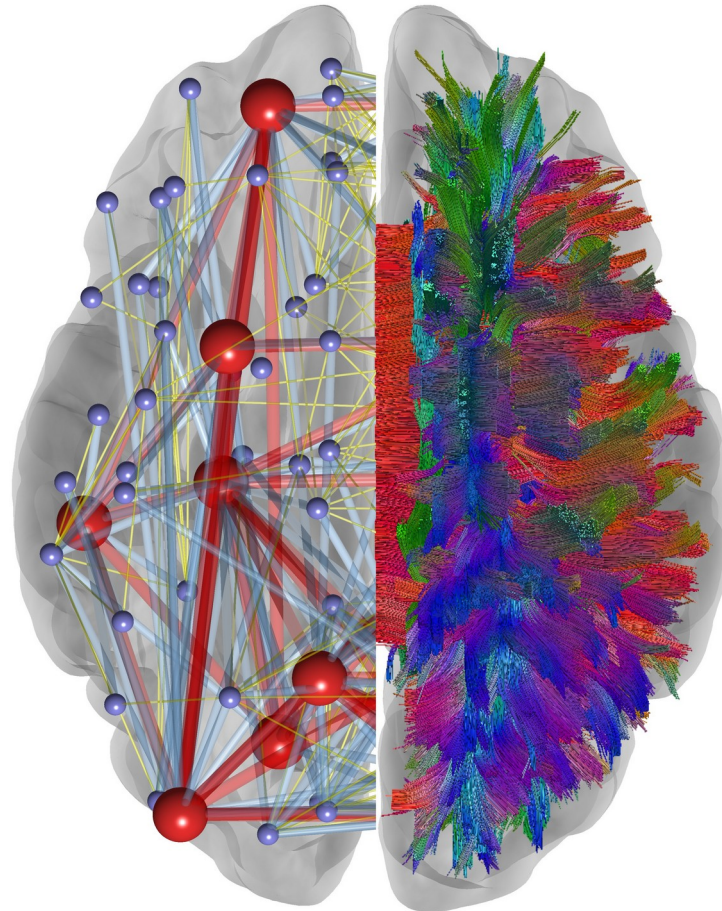
Functional Connectivity x Structural Connectivity

How do neurons communicate with each other?

What is the information flow?

Correlation, phase and Causality

Modifications are task dependent (cognition)



How do neurons connect to each other?

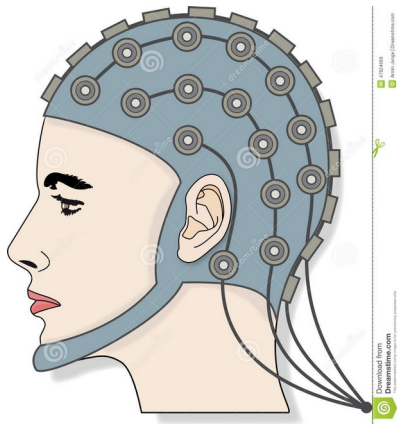
What is the topological organization of anatomical network?

Real synapses

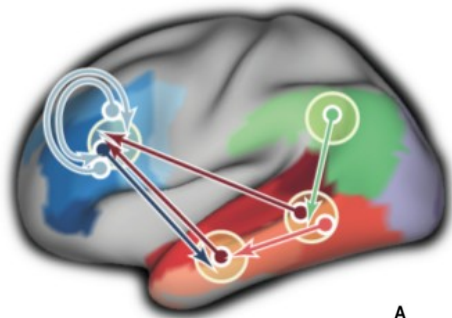
Modifications via synaptic plasticity (STDP)

Main goal: Understand functional connectivity during different tasks and propose neuronal models that can reproduce an experimental observation.

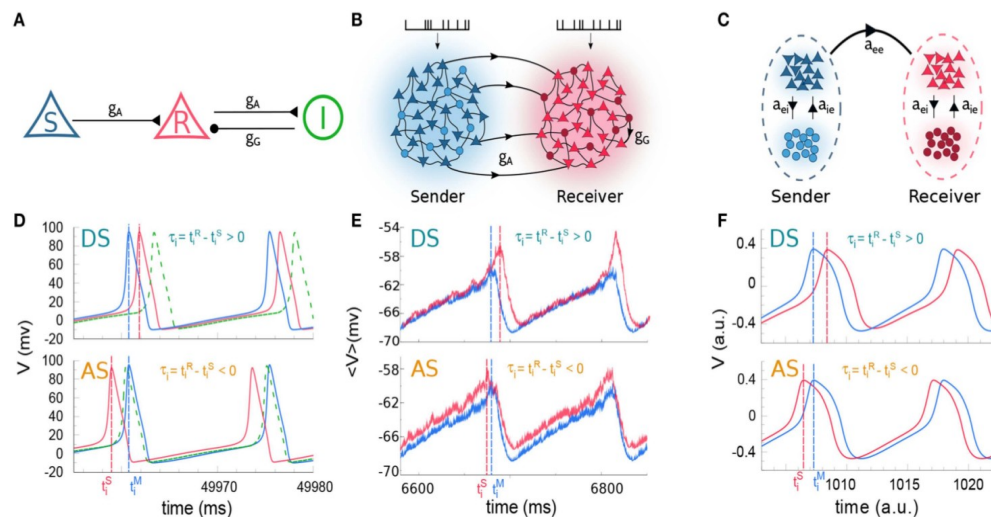
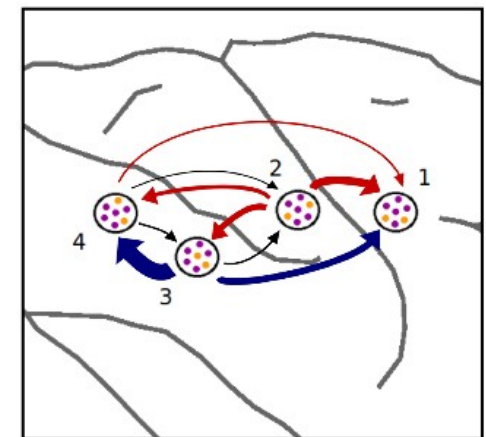
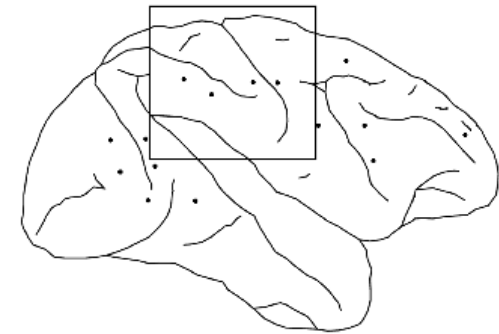
Functional Connectivity x Structural Connectivity



Data analysis: How to infer functional connectivity, information flow, causality, complexity from local field potential (LFP) or electroencephalogram EEG

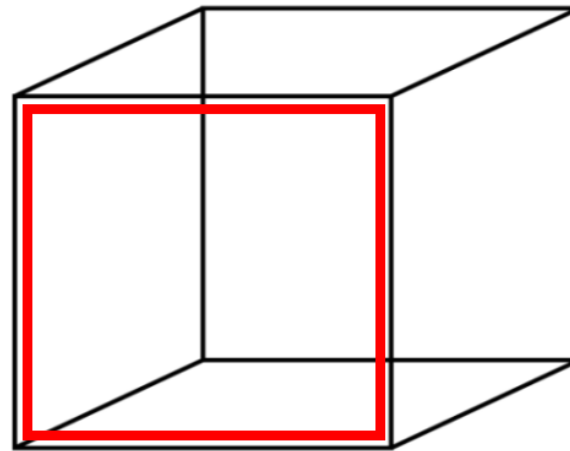
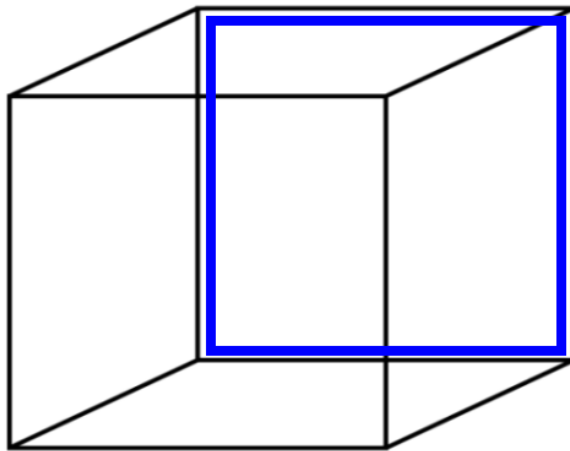


Neuronal simulation: develop brain models in which we know the structural connectivity and we can infer functional connectivity, information flow, causality and complexity as for the real data

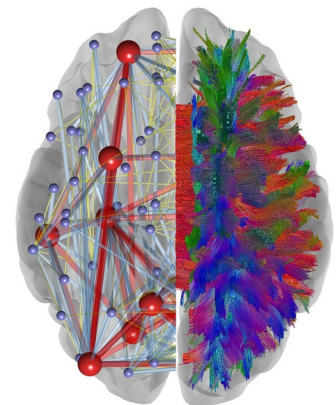


Bistable perception

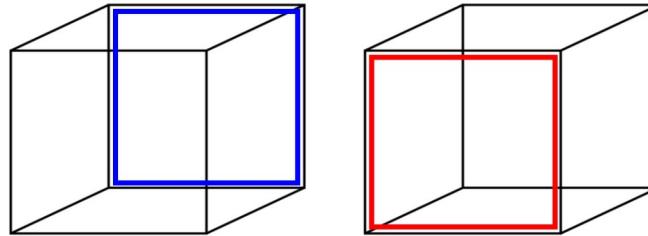
Example 1 (visual): Necker cube



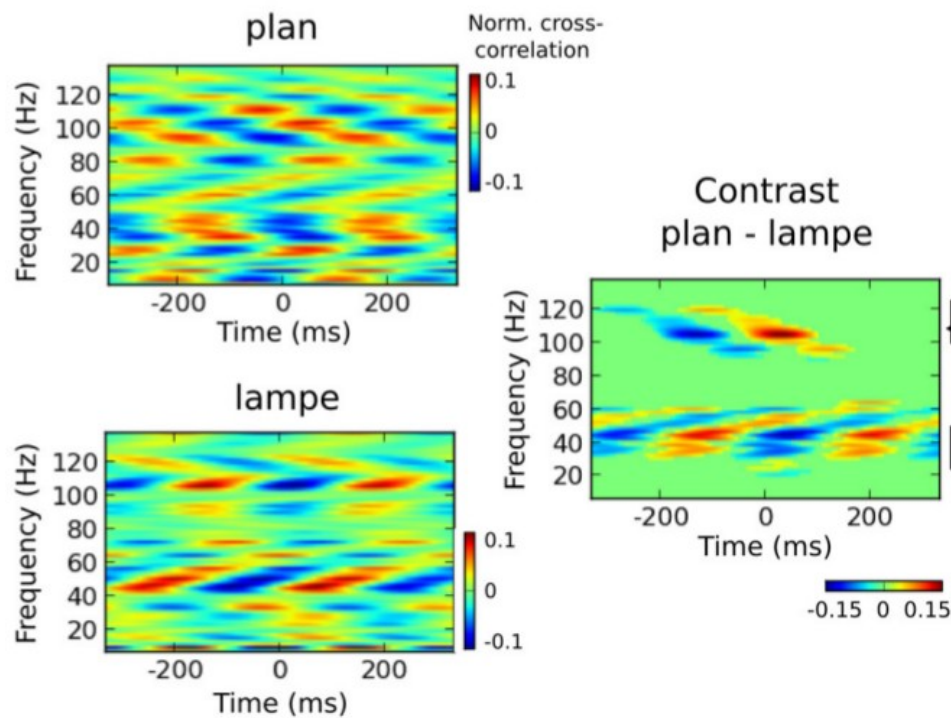
Example 2 (auditive): CA-JA-CA-JA-CA-JA



Bistable perception and Phase bistability

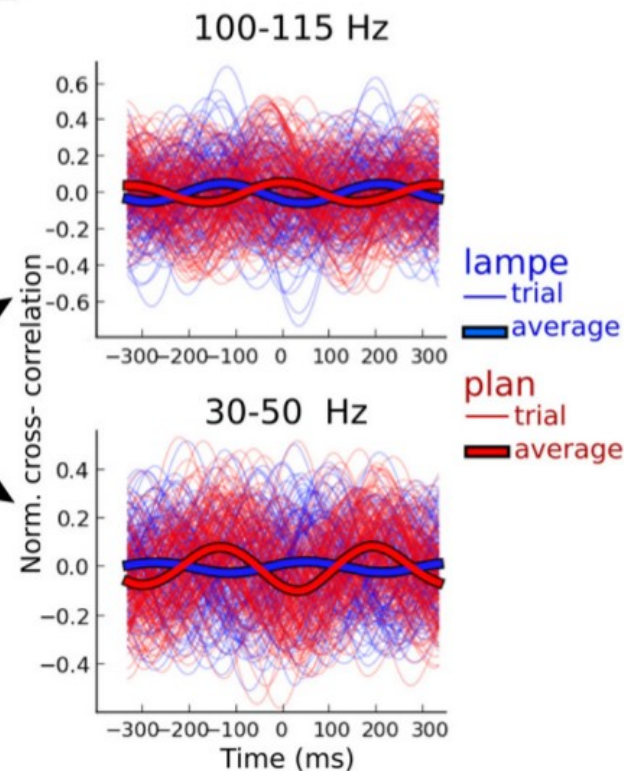


“CAJA”

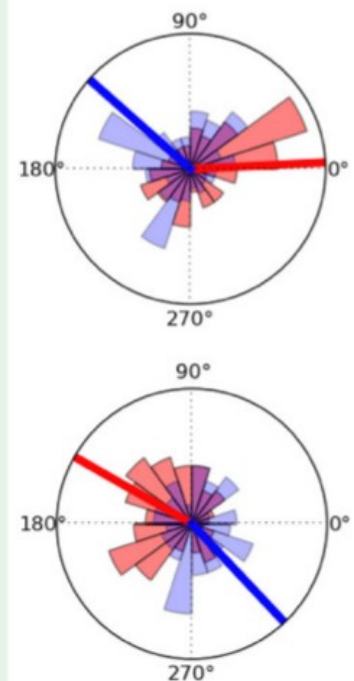


“JACA”

B



C



MEG results suggests that phase relations encode bistable perception (Kosen et al. 2014)

Unidirectional causality with negative phase

Beta oscillations in a large-scale sensorimotor cortical network: Directional influences revealed by Granger causality

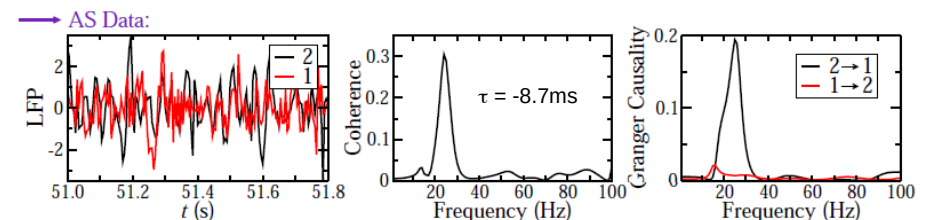
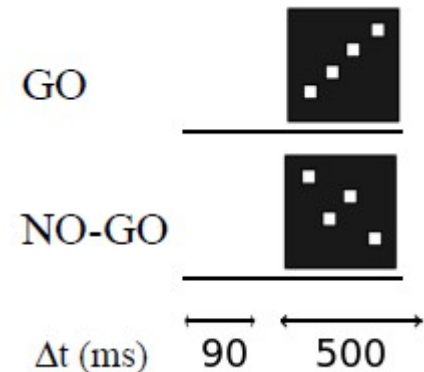
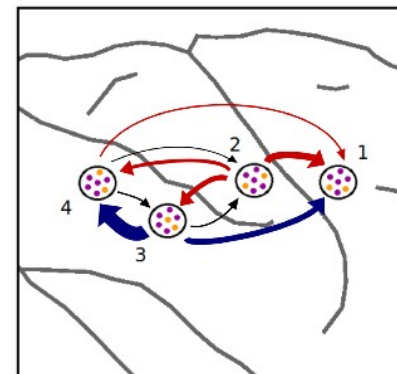
Andrea Brovelli*, Mingzhou Ding*, Anders Ledberg*, Yonghong Chen*, Richard Nakamura†, and Steven L. Bressler**

In general, there was a close correspondence between the characteristics of the Granger causality spectra and the coherence, as suggested by visual comparison of Fig. 1 B and C. All pairs having significant Granger causality beta-range peaks also had significant coherence beta-range peaks. However, Granger causality relations (Tables 3 and 4) were generally inconsistent with time delay values derived from phase spectra (Tables 1 and 2), because the sign of the time delay did not predict the direction of unidirectional Granger causality.

However, no clear relation was found between the time delay derived from the mean phase difference and the Granger causality, thus suggesting that the mean relative phase is not indicative of the directionality of causal relations between cortical assemblies.

Somatosensory G-causes motor cortex.

However, motor area leads in phase.



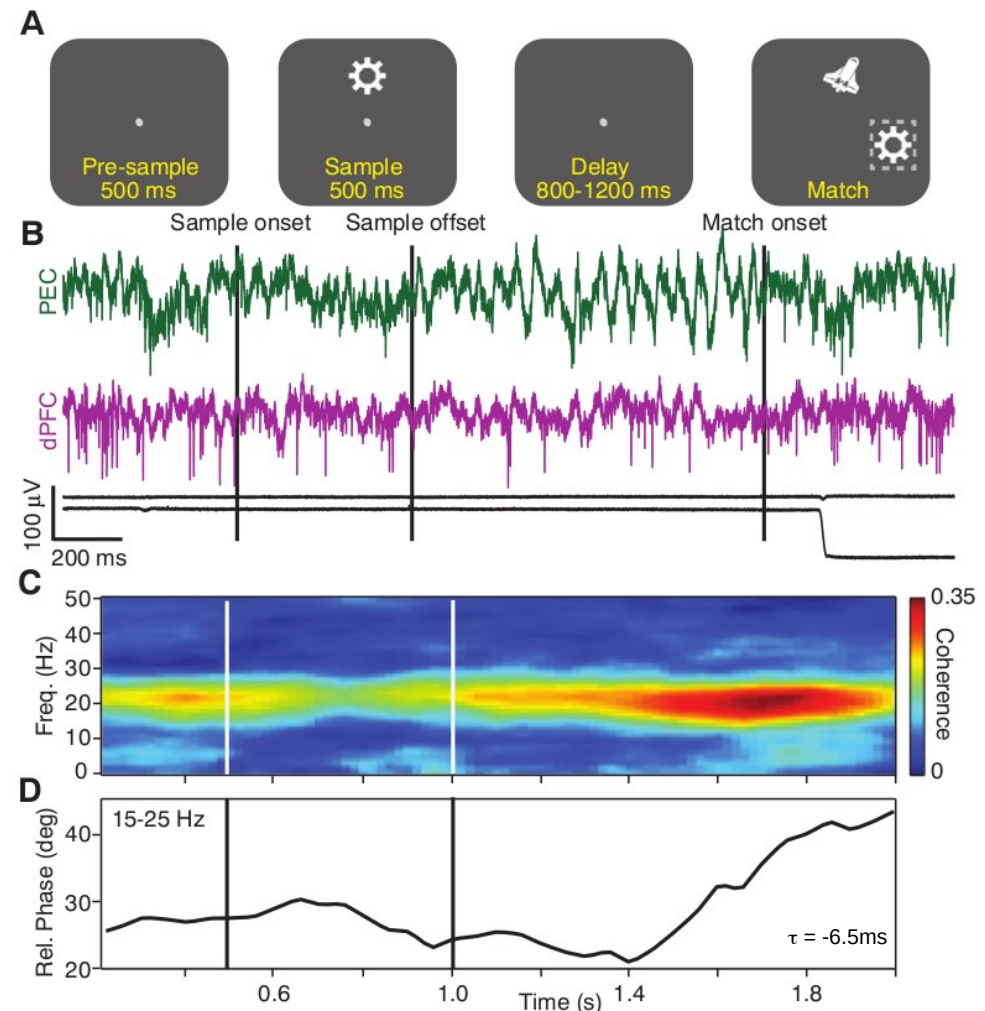
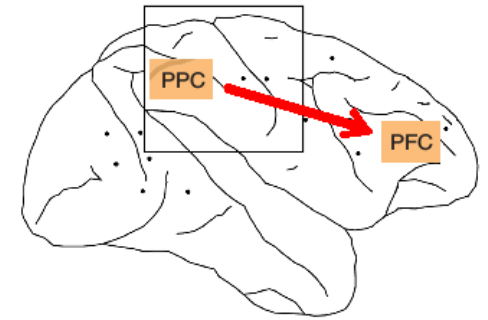
Unidirectional causality with negative phase

Content-Specific Fronto-Parietal Synchronization During Visual Working Memory

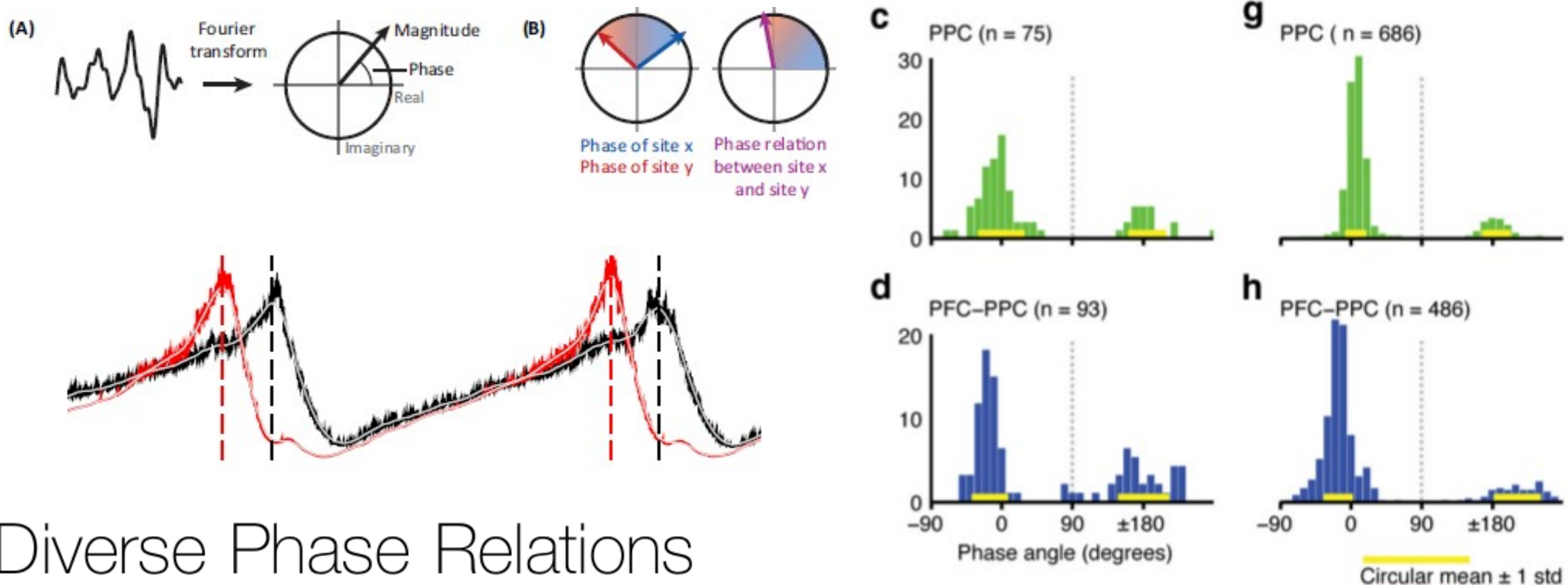
R. F. Salazar,¹ N. M. Dotson,¹ S. L. Bressler,² C. M. Gray^{1*}

The mean relative phase ranged from 15° to 40° throughout the trial (at the center frequency of 17 Hz, these relative phase values correspond to time lags of 2.45 and 6.53 ms, respectively), showing a consistent prefrontal phase lead that gradually increased over time, but showed no

pairs (Fig. 3E, $P < 0.05$, KS test). The agreement between the WGC and SFC results, supporting a dominant PPC to PFC influence, as opposed to the relative phase results showing a PFC phase lead, supports previous findings that relative phase is not a reliable index of neural influence (24).



Phase relation diversity



Diverse Phase Relations among Neuronal Rhythms and Their Potential Function

Eric Maris,^{1,*} Pascal Fries,^{1,2} and Freek van Ede^{1,3}

Rhythmic neuronal synchronization in visual cortex entails spatial phase relation diversity that is modulated by stimulation and attention

Eric Maris ^{a,*}, Thilo Womelsdorf ^{a,b}, Robert Desimone ^{c,d}, Pascal Fries ^{a,e}

Dotson (2014)

Phase diversity could be across space, time, subjects

Let's think about models....

We propose a neuronal model based on **Anticipated Synchronization** ideas in which inhibitory synapses modulate the phase relations

Our models can reproduce many phase relations:

- Unidirectional causality with negative phase
- Shorter time-delays
- Phase relation diversity
- Phase Bistability

What is anticipated synchronization?

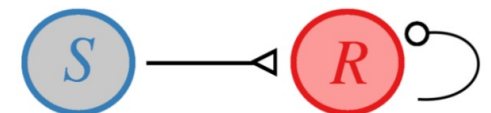
Two identical dynamical systems coupled in a SENDER-RECEIVER configuration can exhibit AS if the receiver also receives a **negative delayed self-feedback**:

$$\begin{aligned}\dot{x} &= f(x(t)), \\ \dot{y} &= f(y(t)) + K[x(t) - y(t - t_d)].\end{aligned}$$

$$y(t) = x(t + t_d) \quad \text{It's a solution!}$$

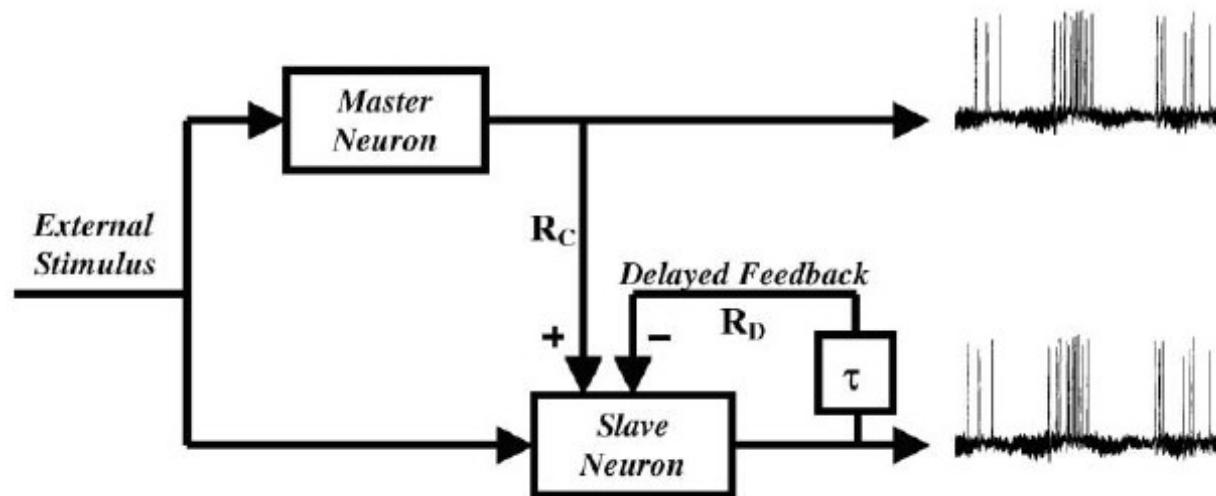
Voss (2000)

No violation of causality



AS between FitzHugh-Nagumo neuron models

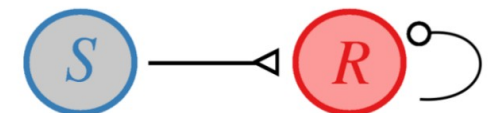
SENDER-RECEIVER configuration with a coupling of the receiver membrane potential with its negative delayed value:



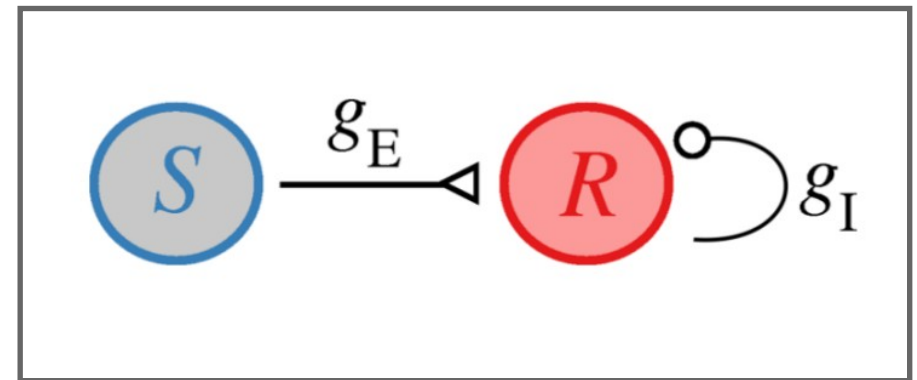
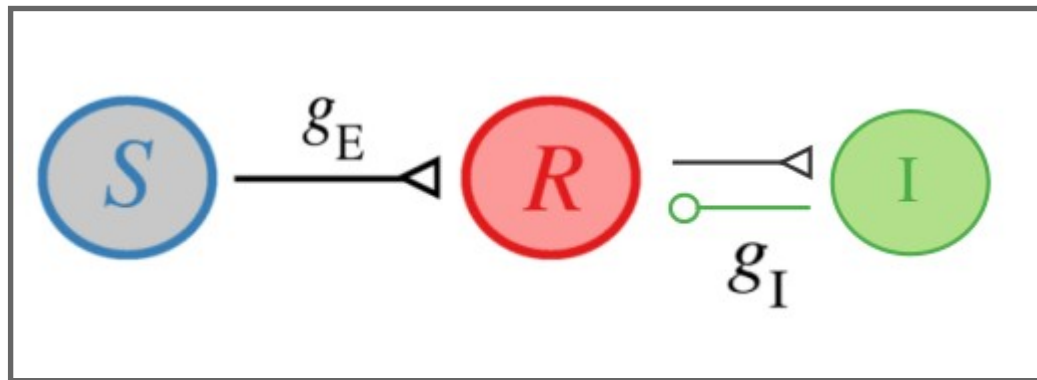
$$\begin{aligned}\dot{x} &= f(x(t)), \\ \dot{y} &= f(y(t)) + K[x(t) - y(t - t_d)].\end{aligned}$$

$$y(t) = x(t + t_d)$$

Ciszak et al. (2003)



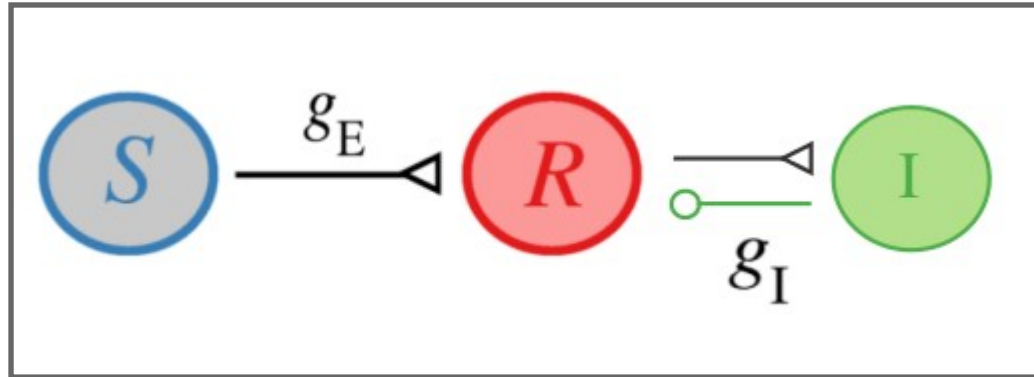
The dynamical inhibitory loop can be mediated by chemical synapse or autapse



$$\begin{aligned}\dot{x} &= f(x(t)), \\ \dot{y} &= f(y(t)) + K[x(t) - y(t - t_d)].\end{aligned}$$

$$y(t) = x(t + t_d)$$

3-Neuron Motif: The Model



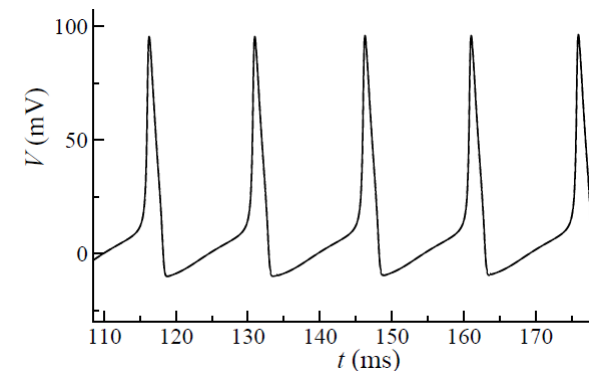
**Hodgkin-Huxley
Neuron Model**

$$C_m \frac{dV}{dt} = \overline{G}_{Na} m^3 h (E_{Na} - V) + \overline{G}_K n^4 (E_K - V) \quad \text{Membrane potential} \\ + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

**Constant external
current**

$$I = 280 \text{ pA} \quad T = 14.69 \text{ ms}$$



**Synaptic current
AMPA and GABAA**

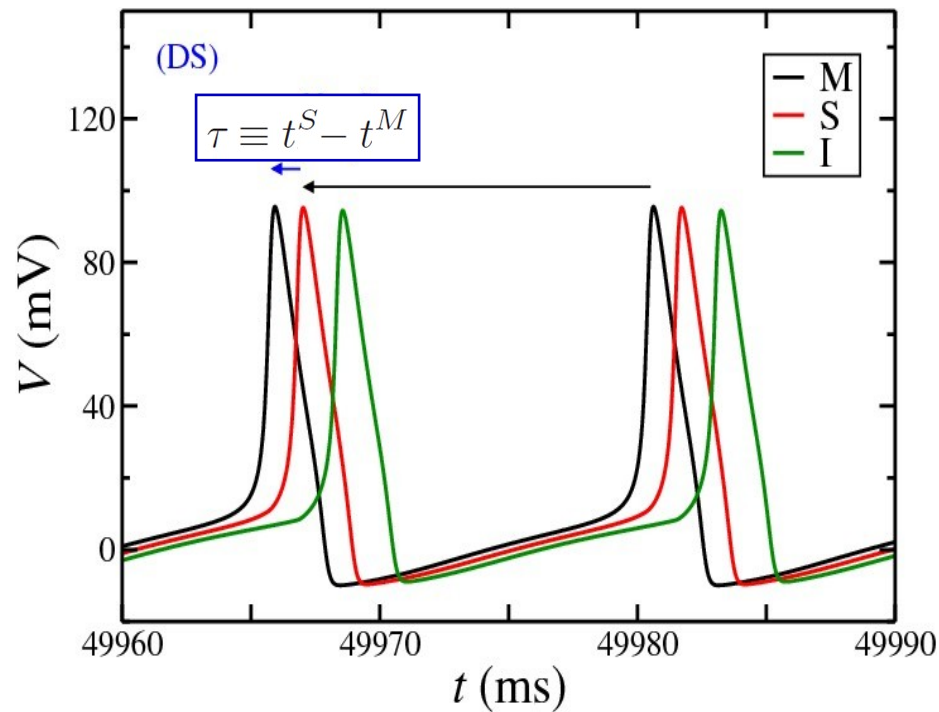
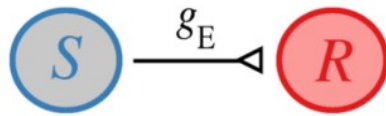
$$I^{(i)} = g_i r^{(i)} (V - E_i)$$

$$\frac{dr^{(i)}}{dt} = \alpha_i [T] (1 - r^{(i)}) - \beta_i r^{(i)}$$

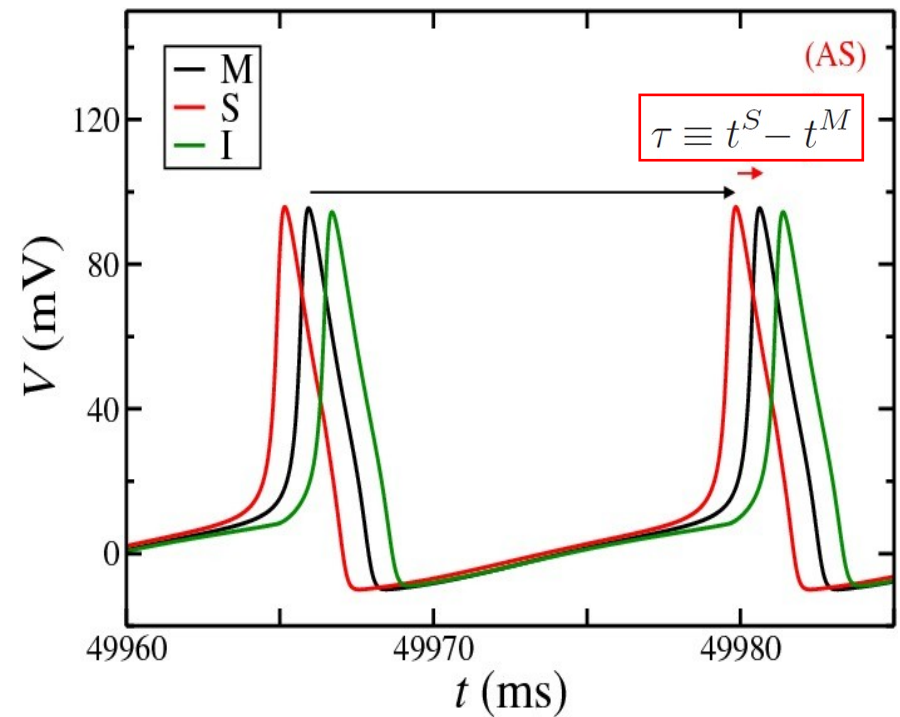
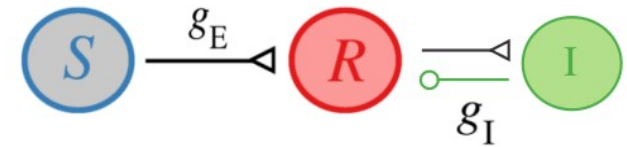
$$I_{syn}(t) = g_{syn} V_{syn} \sum_{\text{spikes}} \alpha(t - t_{\text{spike}})$$

$$\alpha(t) = \pm \frac{1}{\tau_- - \tau_+} [\exp(-t/\tau_-) - \exp(-t/\tau_+)]$$

Anticipates synchronization in neuronal microcircuits



Delayed Synchronization (DS)
master spikes “prior” to slave



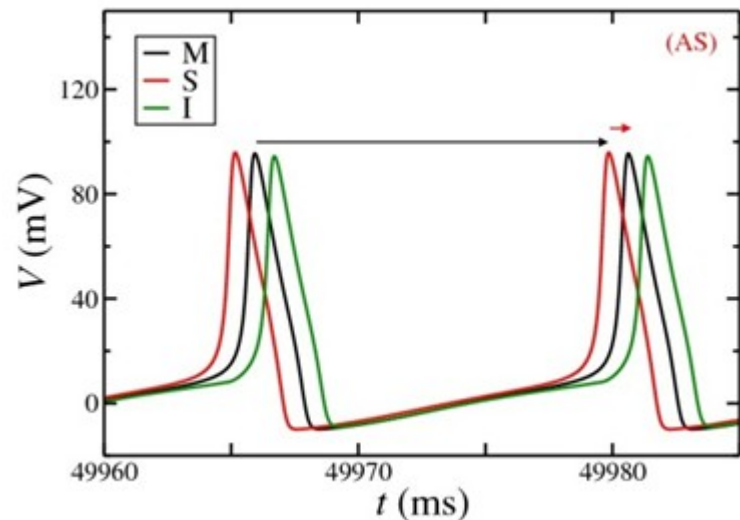
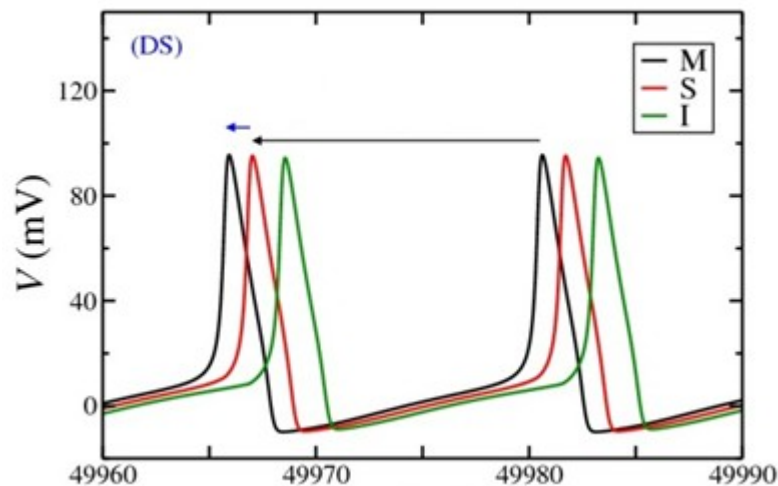
Anticipated Synchronization (AS)
slave spikes “prior” to master

We define the Time Delay (τ):

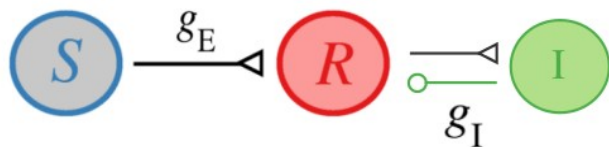
$$\tau \equiv t_i^S - t_i^M$$

Time Delay vs Inhibitory Conductance

Delayed-Synchronization (DS)



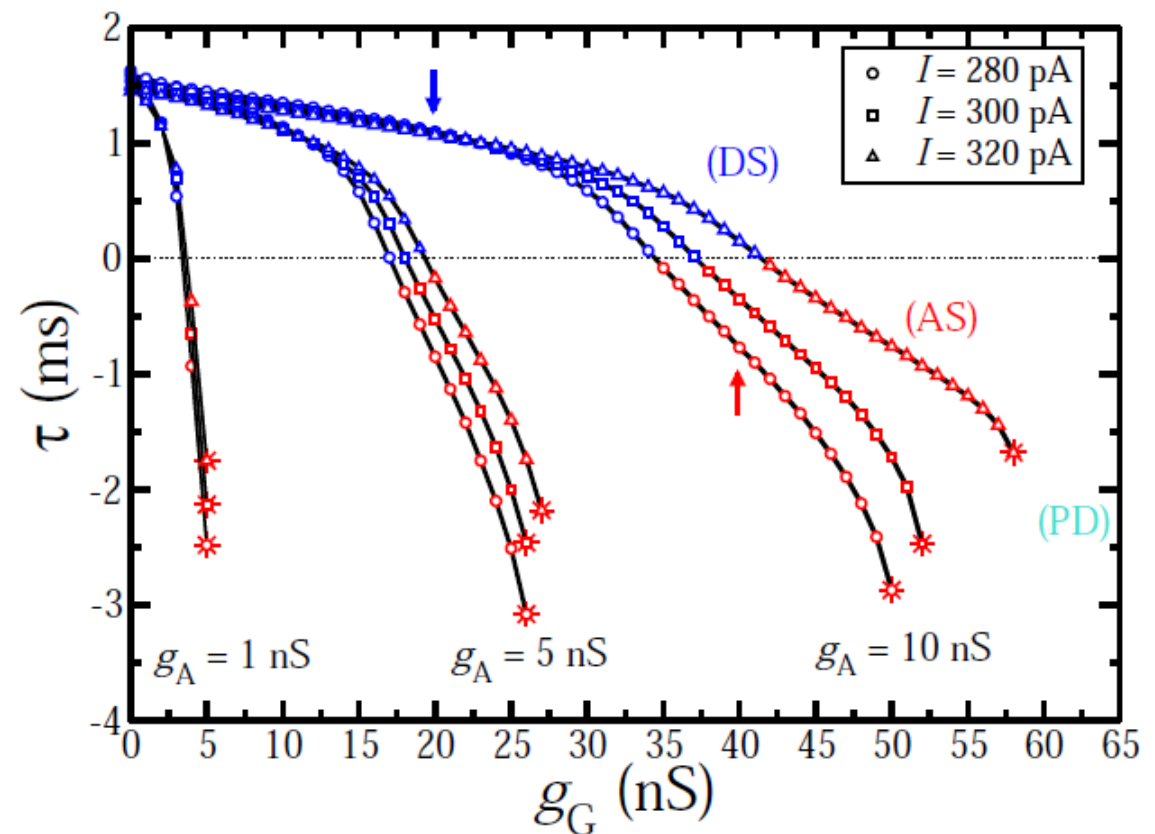
Anticipated-Synchronization (AS)



$$\tau_i \equiv t_i^S - t_i^M$$

$\tau > 0$ (positive) DS

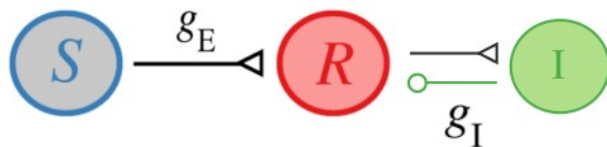
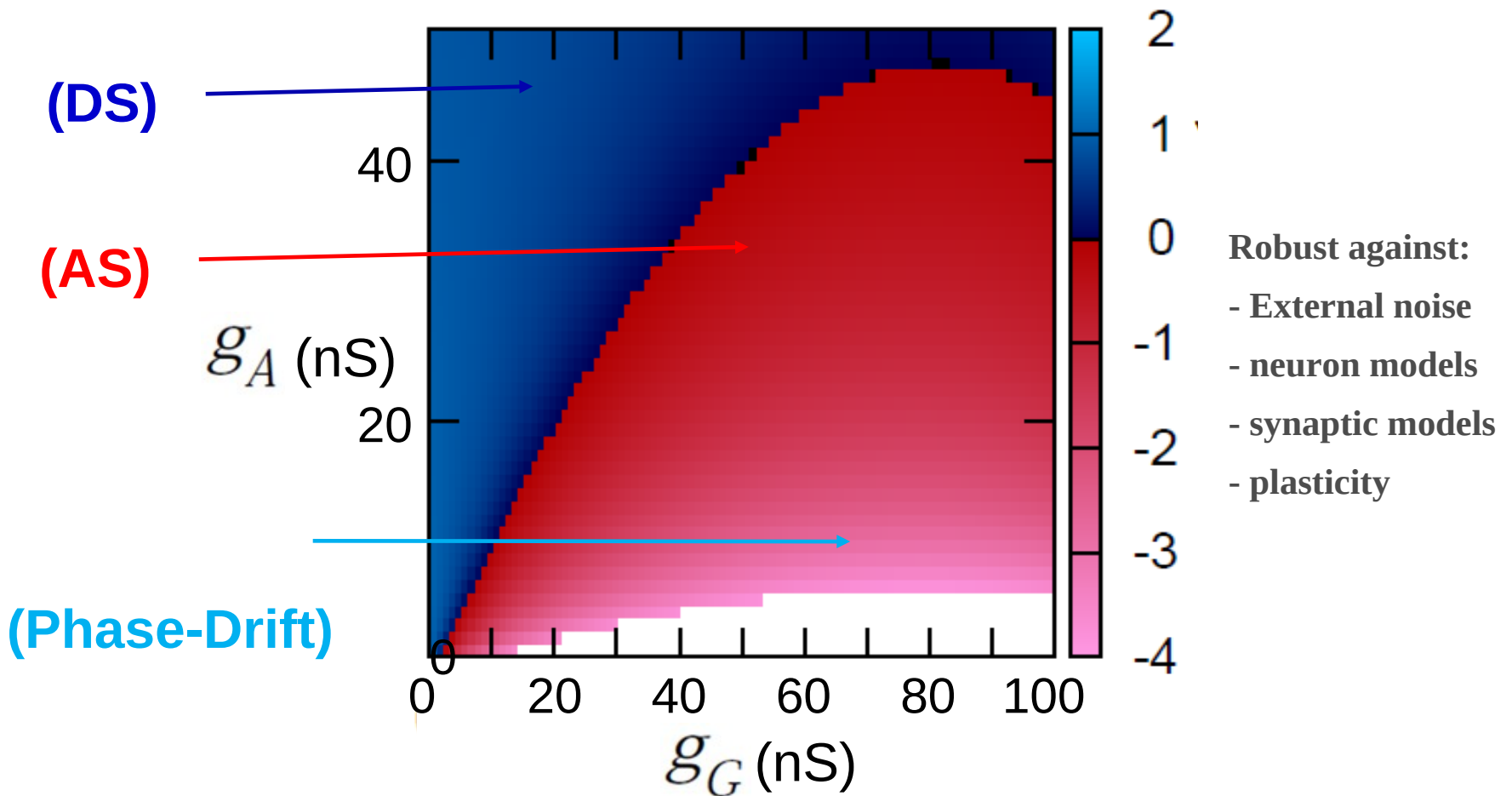
$\tau < 0$ (negative) AS



For large values of g_G the system exhibits a phase-drift (PD) regime.

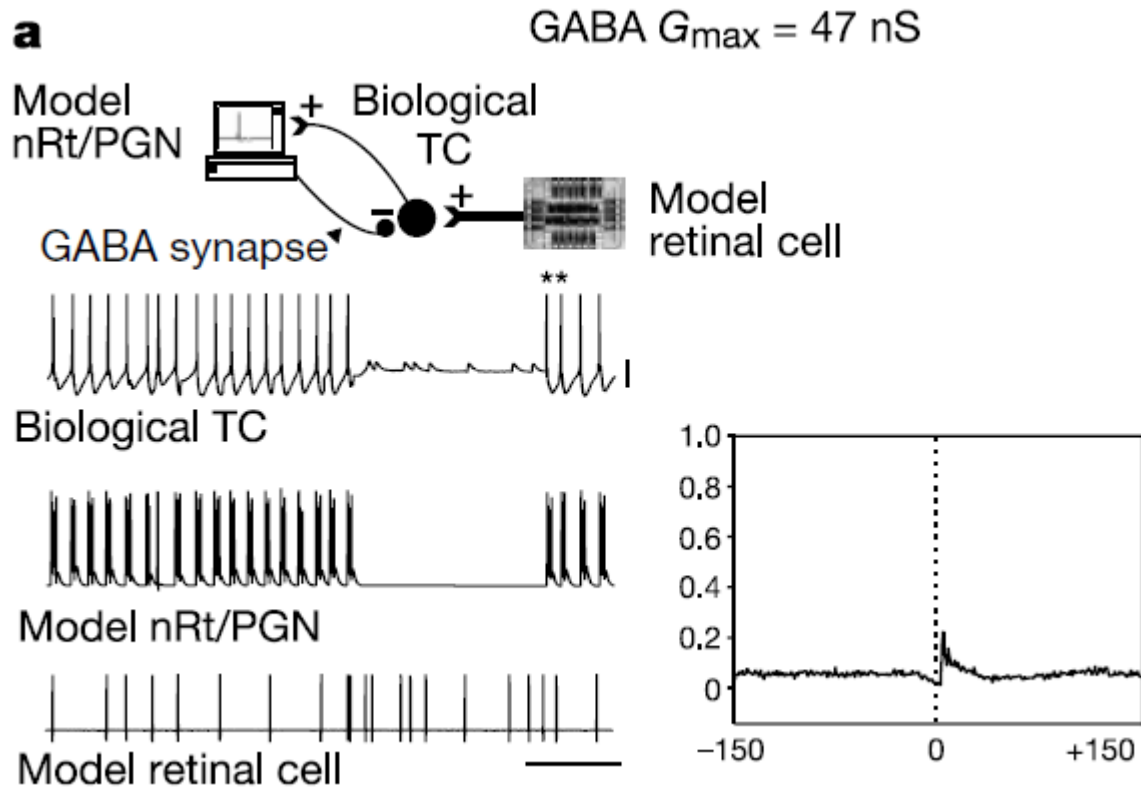
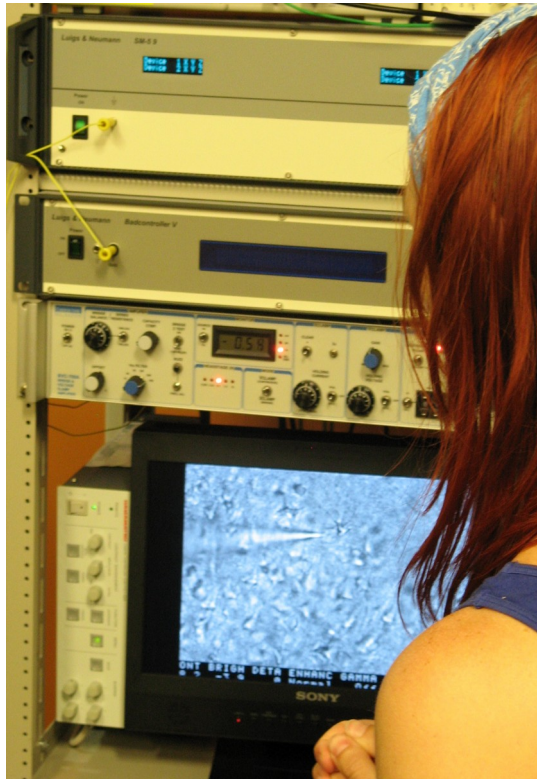
Time Delay vs Excitatory and Inhibitory Conductances

Large regions of AS and DS in the parameter space



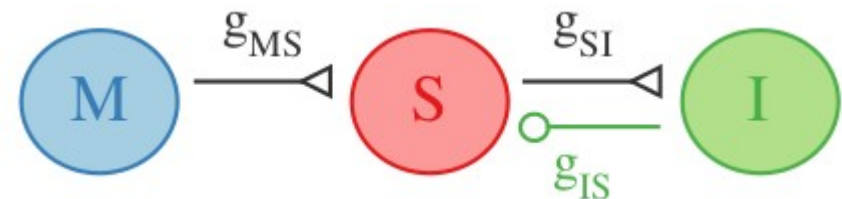
Can we use a phase response curve to predict the phase-differences?

Results can be tested in a hybrid patch clamp setup

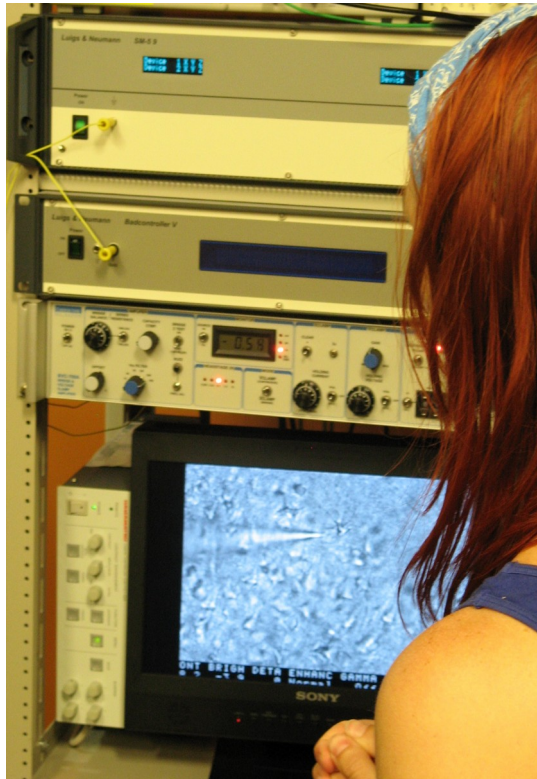


Feedback inhibition controls spike transfer in hybrid thalamic circuits

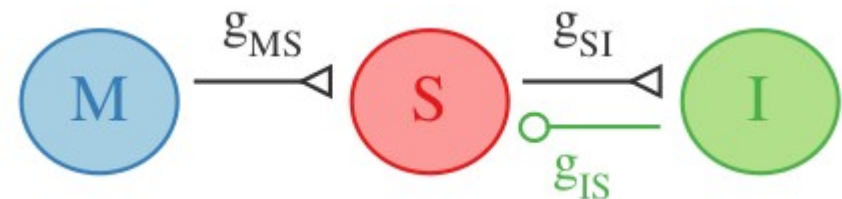
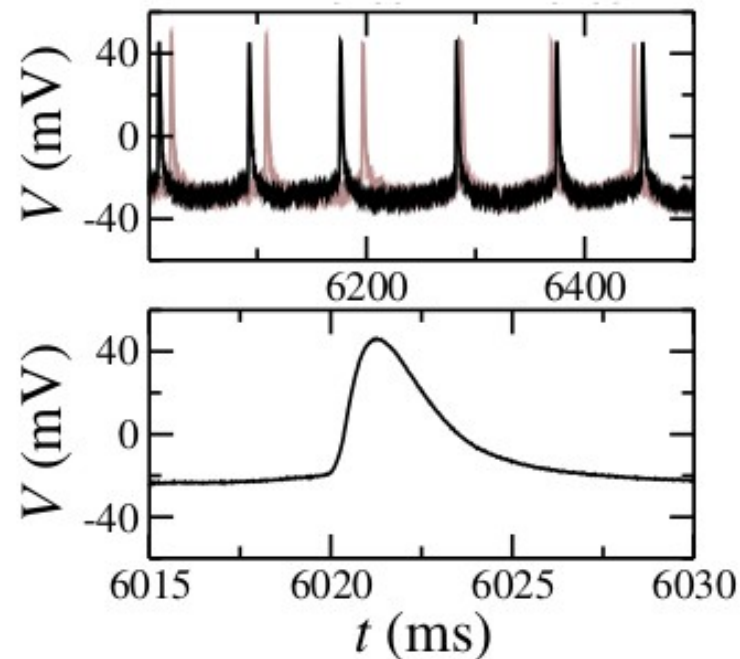
Gwendal Le Masson*, Sylvie Renaud-Le Masson†, Damien Debay‡
& Thierry Bal‡



Results can be tested in a hybrid patch clamp setup



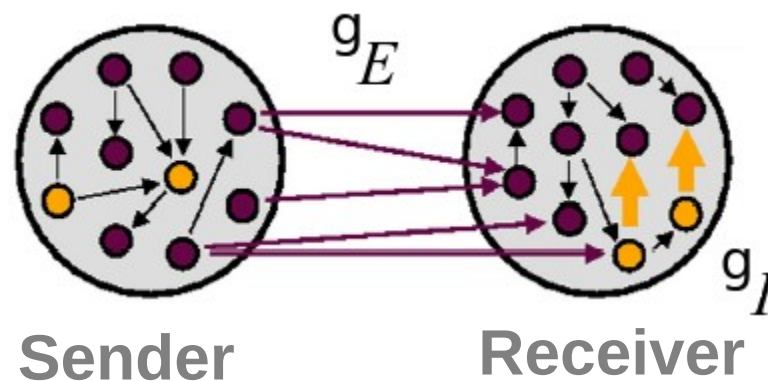
$$C_m \frac{dV}{dt} = \bar{G}_{Na} m^3 h (E_{Na} - V) + \bar{G}_K n^4 (E_K - V) + \bar{G}_{MP}^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$



Cortical-like neuronal population

Cortical-like neuronal population

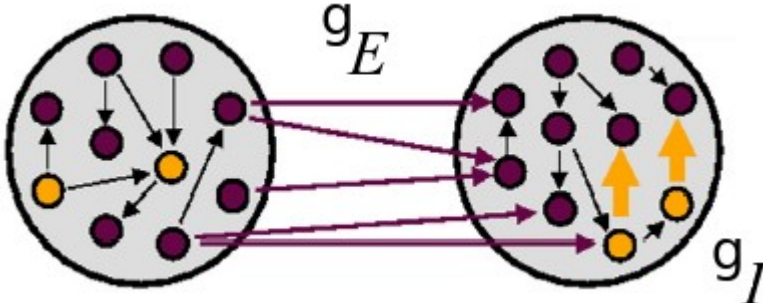
We have replaced the **negative delayed self-feedback** by the internal inhibition in the Receiver population



$$\begin{aligned}\dot{x} &= f(x(t)), \\ \dot{y} &= f(y(t)) + K[x(t) - y(t - t_d)].\end{aligned}$$

$$y(t) = x(t + t_d)$$

Neuronal population model



Izhikevich Neuron Model (neuronal variability)

$$\begin{aligned}\frac{dv}{dt} &= 0.04v^2 + 5v + 140 - u + \sum I_x, \\ \frac{du}{dt} &= a(bV - u).\end{aligned}$$

Synaptic current AMPA and GABAA

$$I_x = g_x r_x (v - E_x)$$

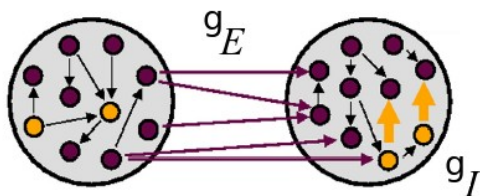
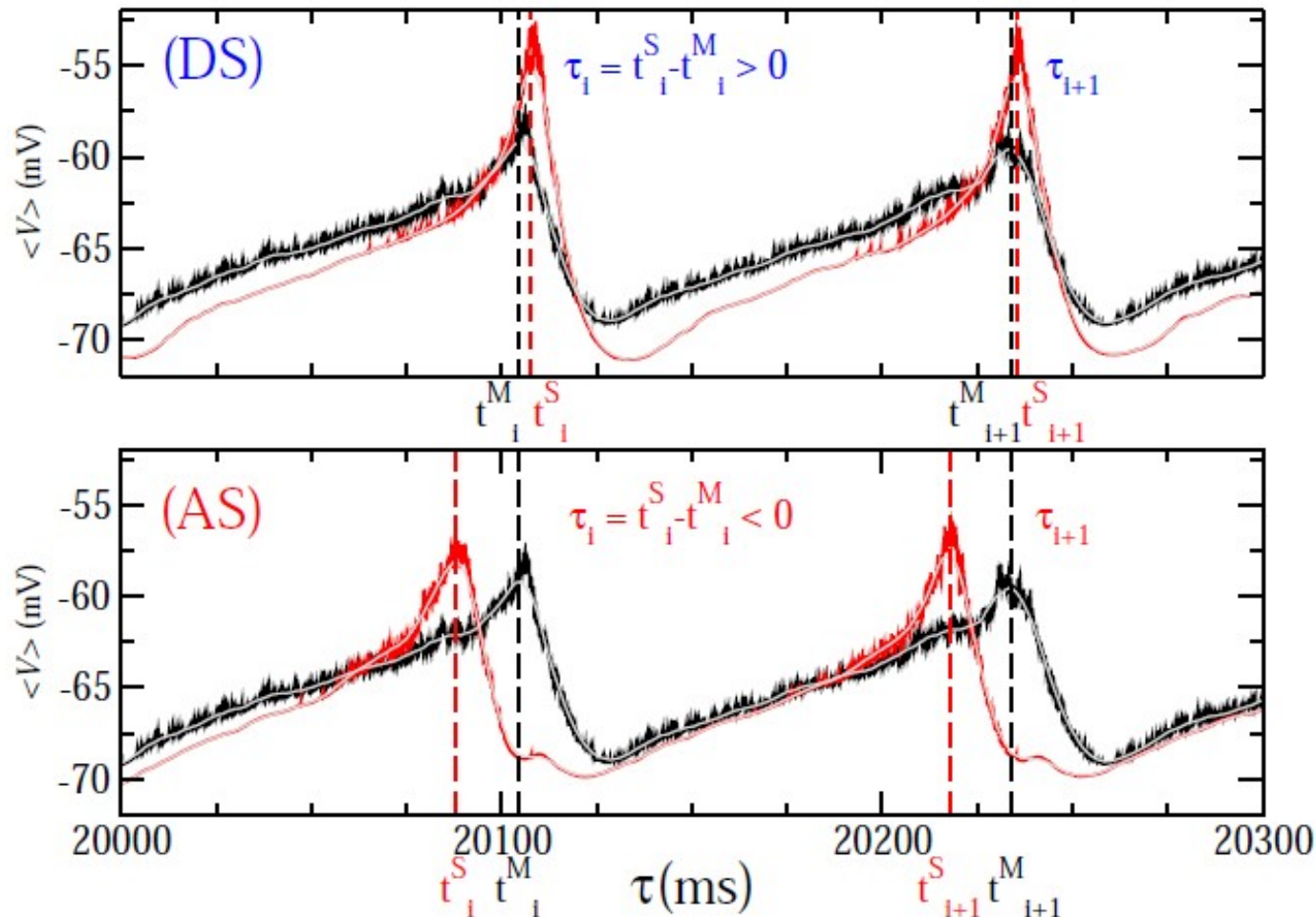
$$\tau_x \frac{dr_x}{dt} = -r_x + \sum_k \delta(t - t_k)$$

Each neuron receive an independent Poison input (noise)

Mean membrane potential

$$\tau_i \equiv t_i^S - t_i^M$$

Anticipated synchronization in neuronal populations!

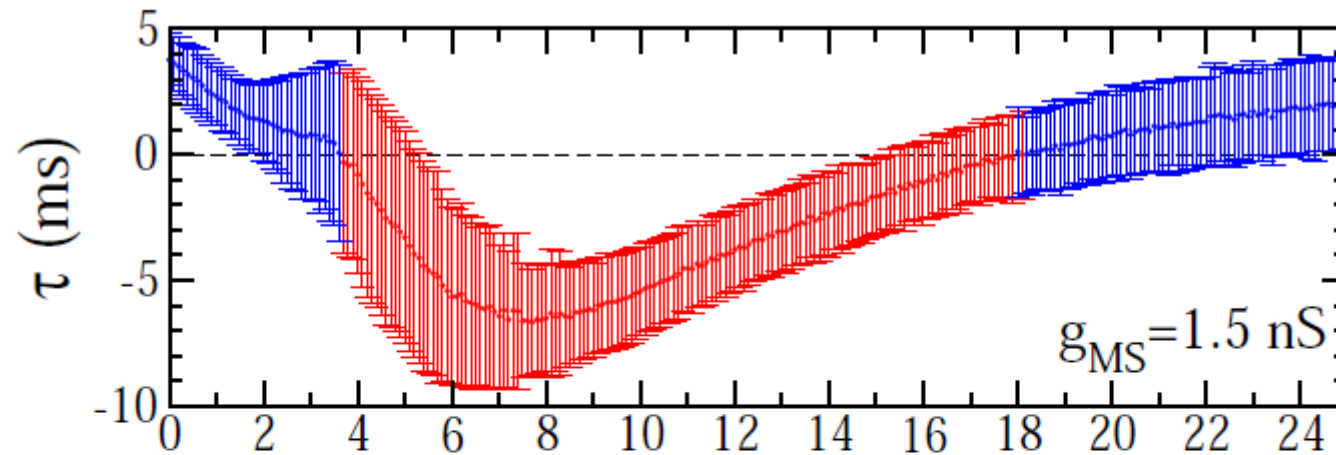


Mean Period $T = 130$ ms ($f = 7.7$ Hz)

Time delay is a function of the inhibitory conductance

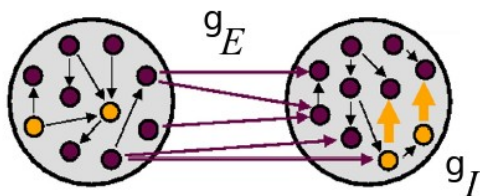
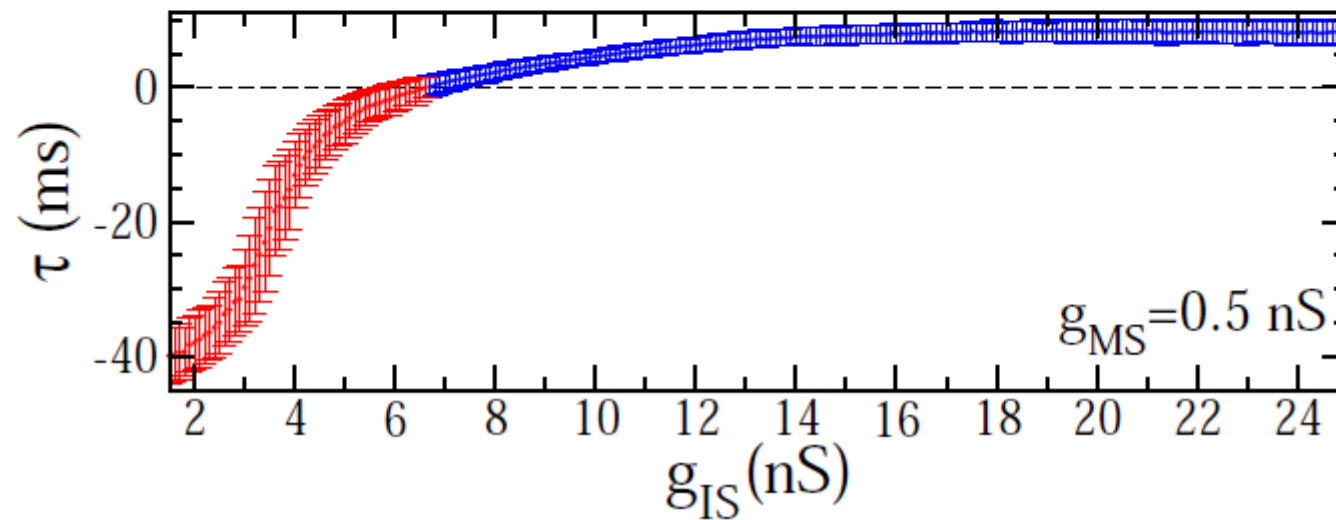
(DS)

(AS)

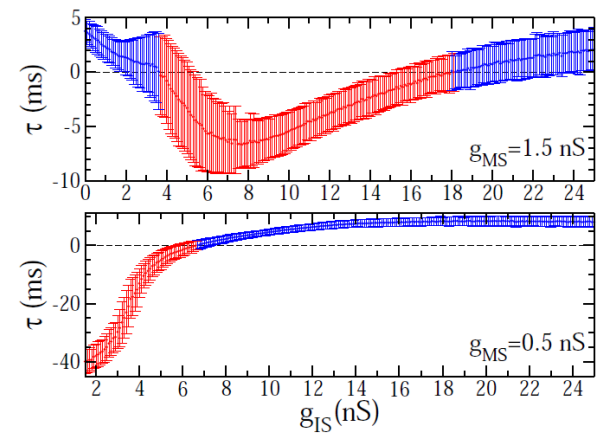
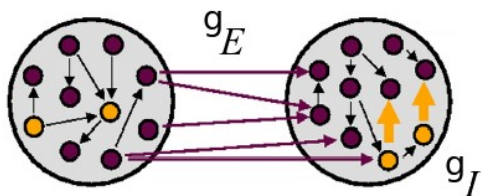
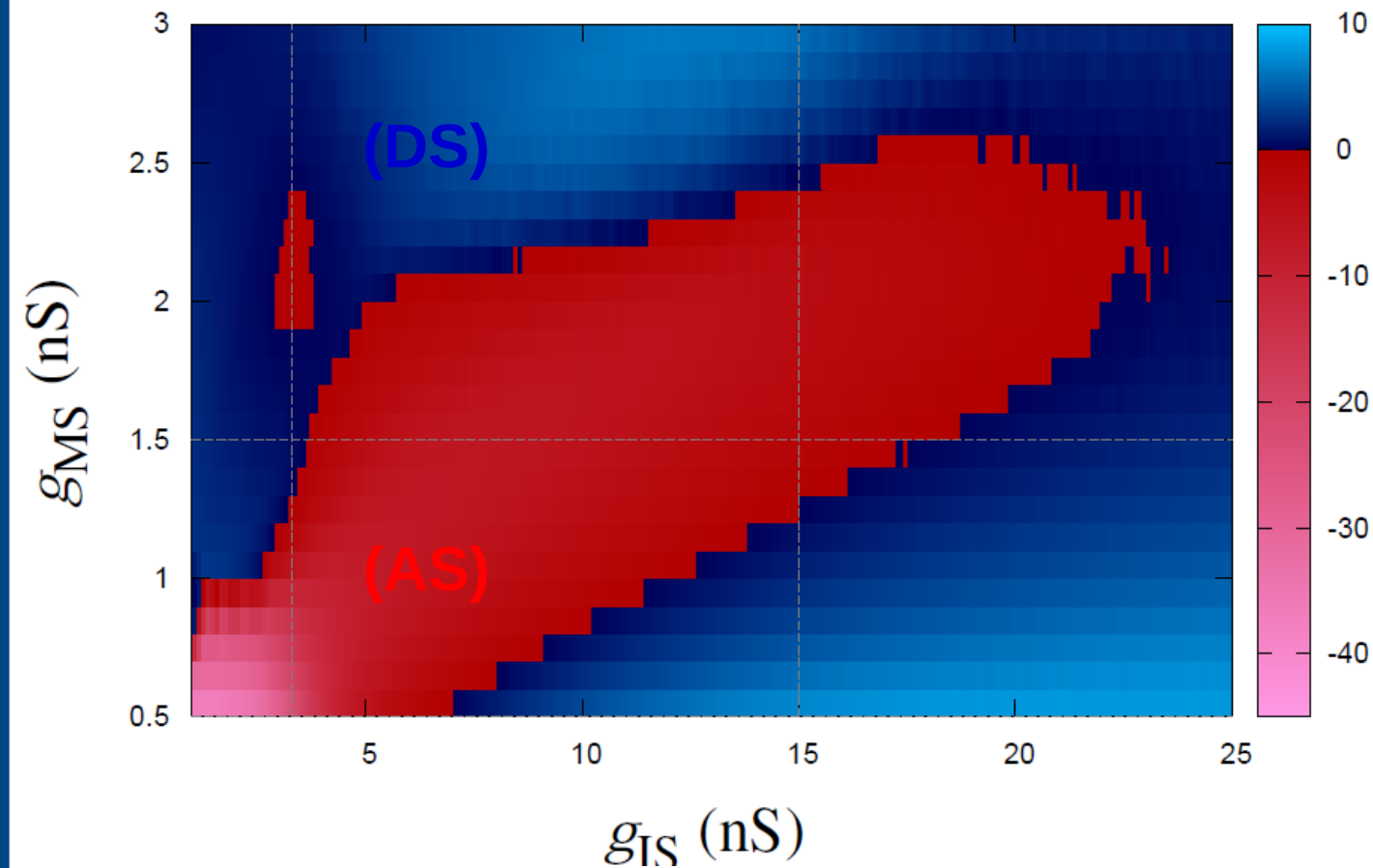


(DS)

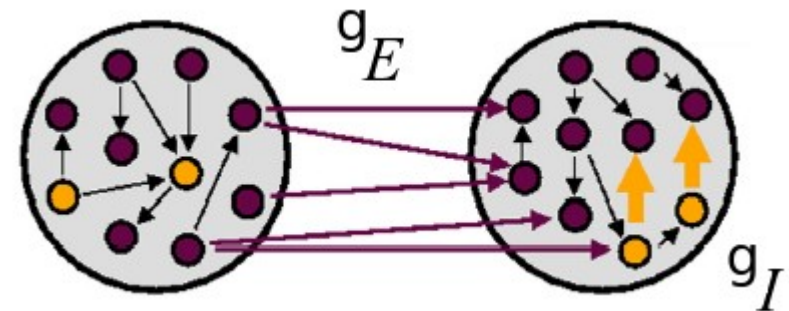
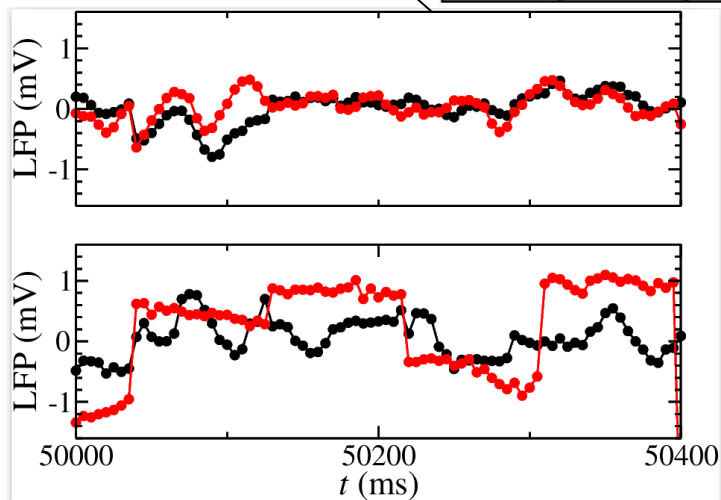
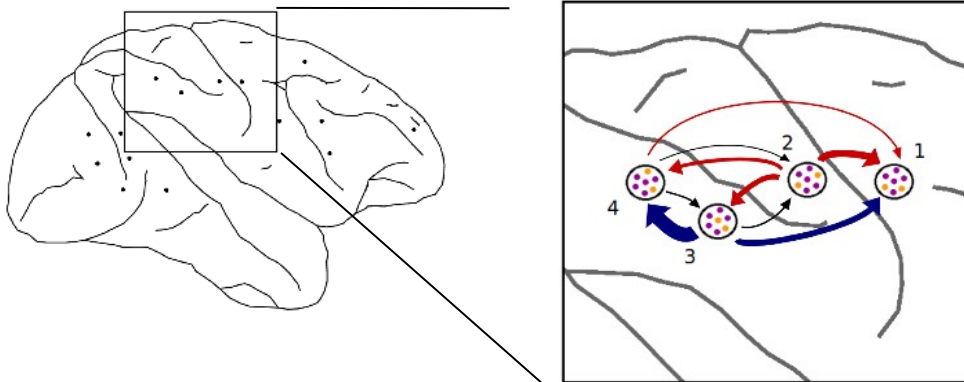
(AS)



Time delay as a function of the conductances



Comparing data and model: Granger causality

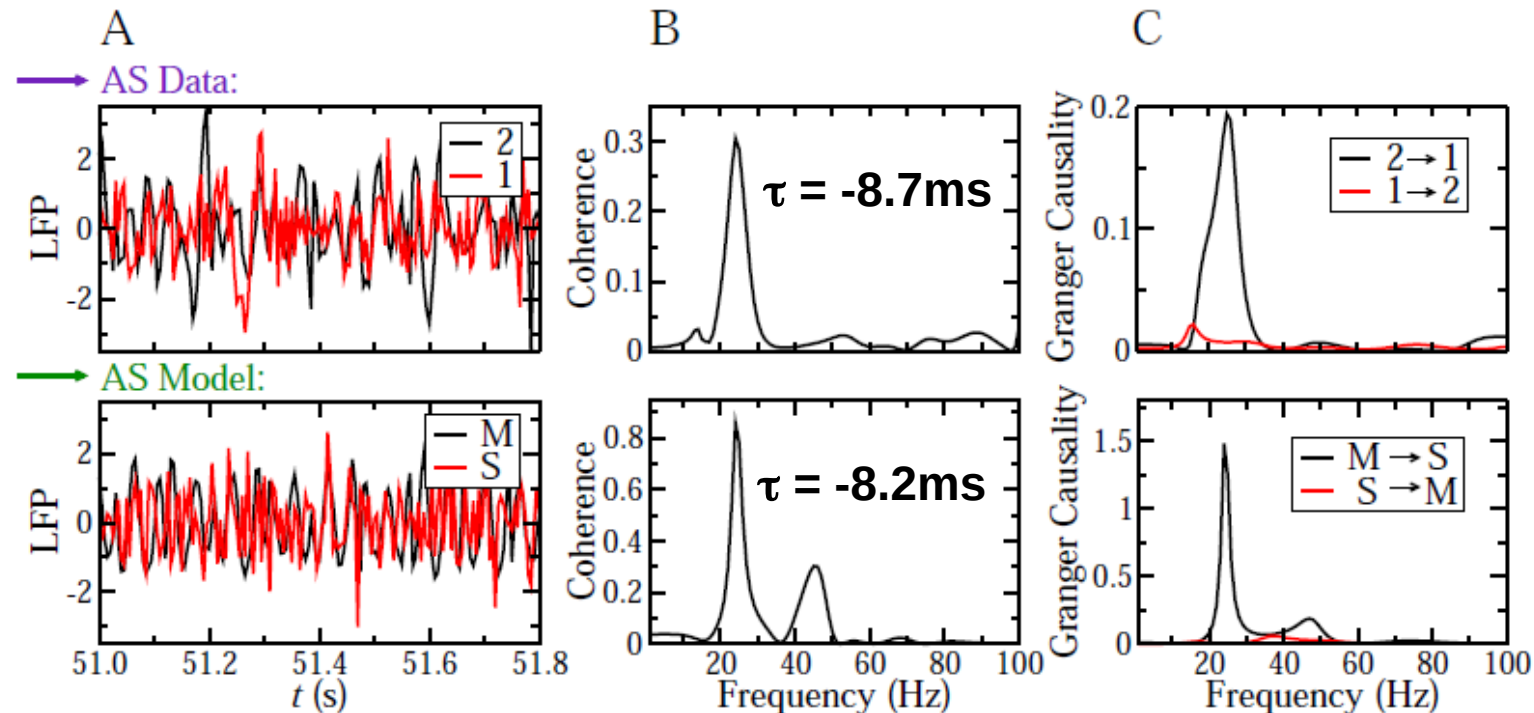


Considering a bivariate linear autoregressive model of two variables X_1 and X_2 . If the inclusion of X_2 decreases the variance of E_1 (A_{12} is different from zero), this means that X_2 helps to forecast X_1 and we say X_2 Granger causes X_1

$$X_1(t) = \sum_{j=1}^p A_{11,j} X_1(t-j) + \sum_{j=1}^p A_{12,j} X_2(t-j) + E_1(t)$$

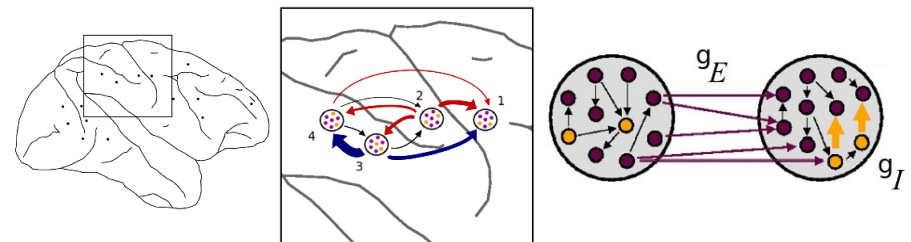
$$X_2(t) = \sum_{j=1}^p A_{21,j} X_1(t-j) + \sum_{j=1}^p A_{22,j} X_2(t-j) + E_2(t)$$

Anticipated synchronization in neuronal data: unidirectional causality with negative phase lag



Somatosensory cortex G-causes motor cortex.

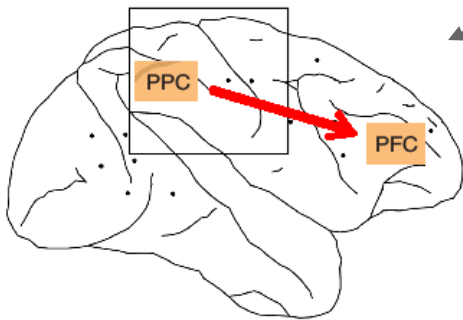
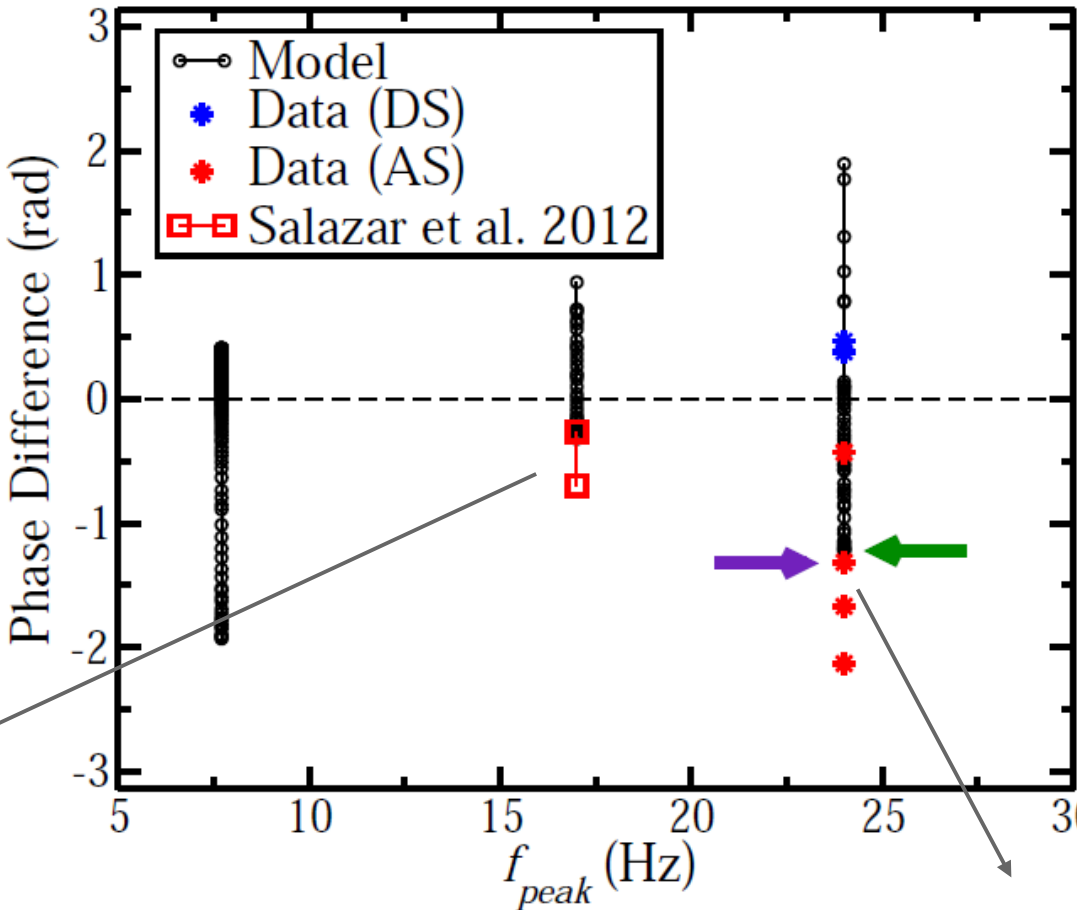
However, motor area leads in phase.



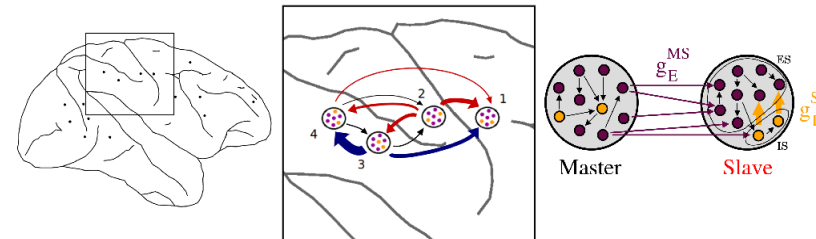
Electrophysiological data may exhibit unidirectional causality with negative lag-synchronization during oscillatory periods (anticipated synchronization).

Matias et al. Neuroimage (2014)

Comparing DATA and MODEL



Electrophysiological data may exhibit unidirectional causality with negative lag-synchronization during oscillatory periods (anticipated synchronization).



Matias et al. Neuroimage (2014)

Phase diversity in EEG data

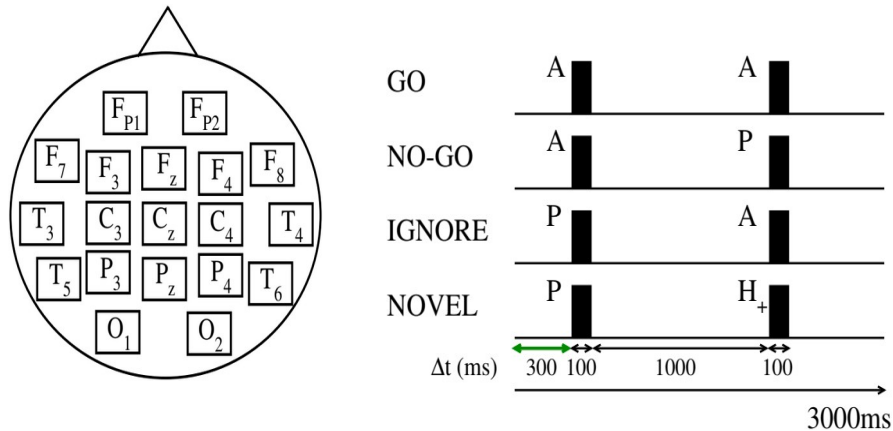
AS in human EEG

PHYSICAL REVIEW E **102**, 032216 (2020)

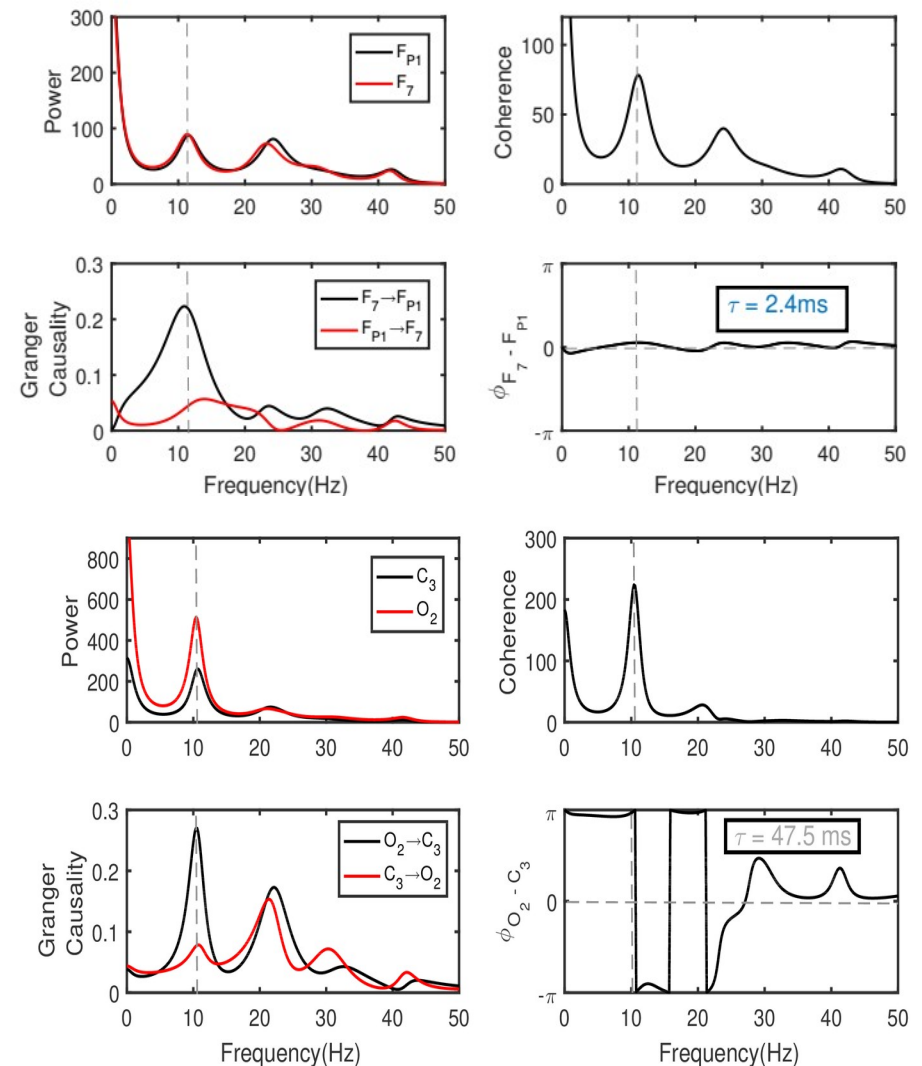
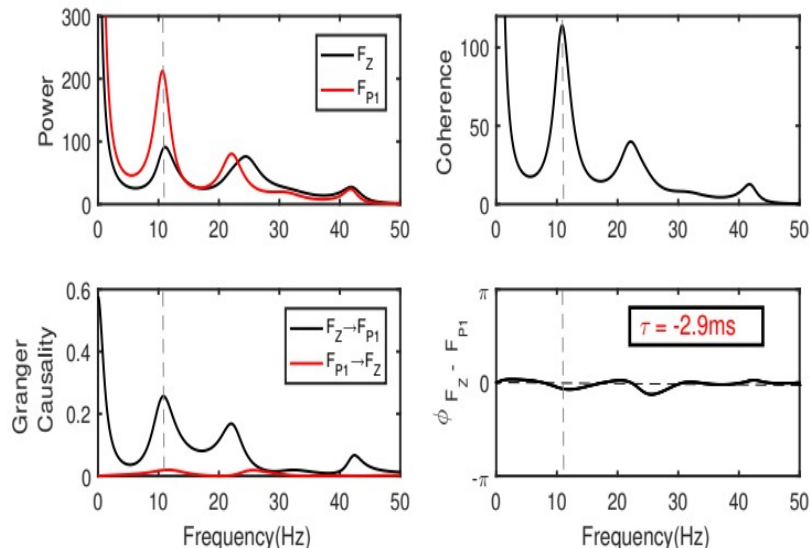
→ Looking for AS in human EEG during a GO/NO-GO task:

Anticipated synchronization in human EEG data: Unidirectional causality with negative phase lag

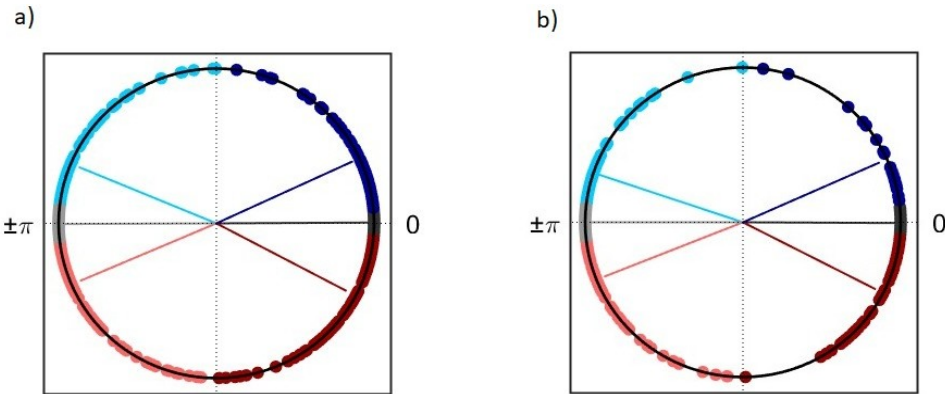
Francisco-Leandro P. Carlos¹, Maciel-Monteiro Ubirakitan^{2,3}, Marcelo Cairrão Araújo Rodrigues²,
Moisés Aguilar-Domingo^{3,4}, Eva Herrera-Gutiérrez⁵, Jesús Gómez-Amor^{4,*}, Mauro Copelli⁶,
Pedro V. Carelli⁶ and Fernanda S. Matias^{1,†}



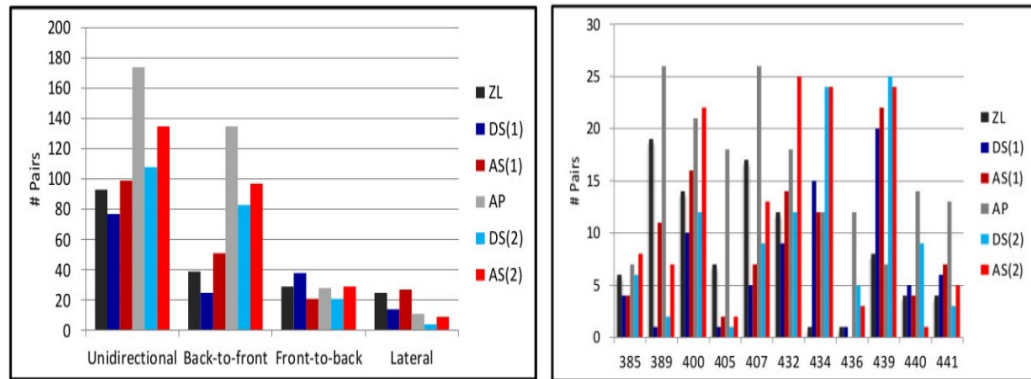
→ AS between EEG channels: unidirectional causality with negative phase-lag:



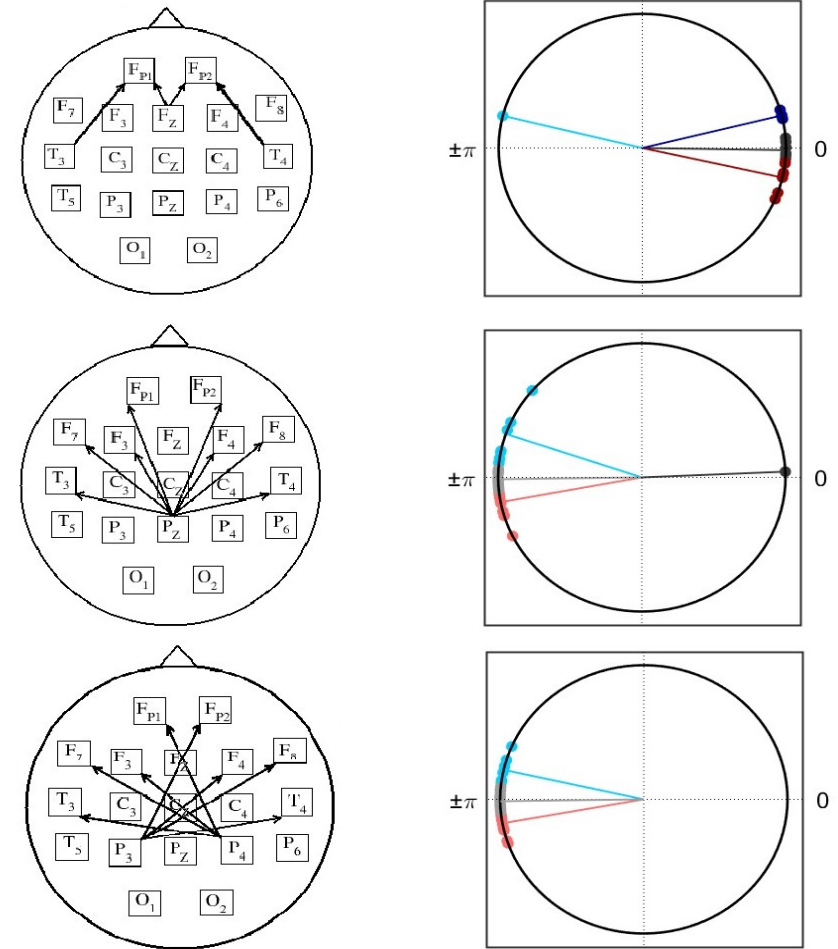
Phase diversity in human EEG



Circular phase differences distribution. Phase of all 686 unidirectionally connected pairs: (b) 430 pairs showing back-to-front influence.



(a) Electrode pairs are separated by direction of influence: all unidirectional pairs, back-to-front influence, front-to-back and lateral direction. (b) All unidirectional pairs separated per volunteer.



Phase bistability

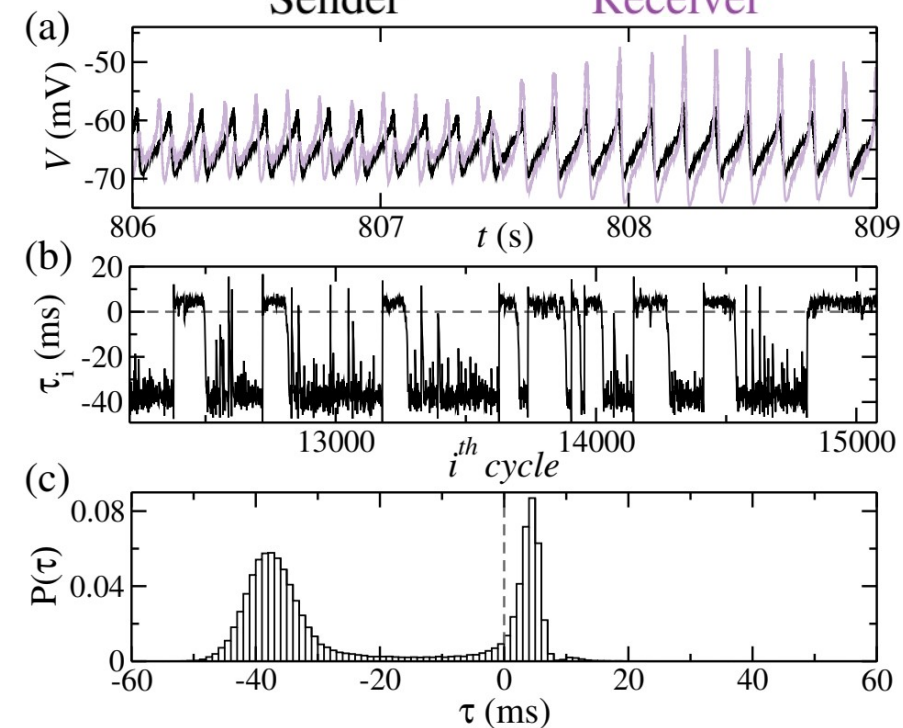
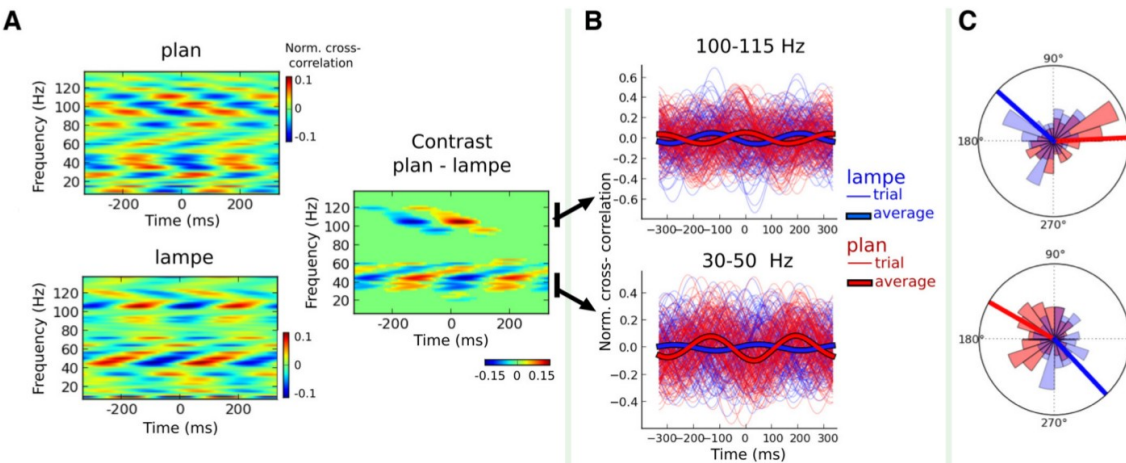
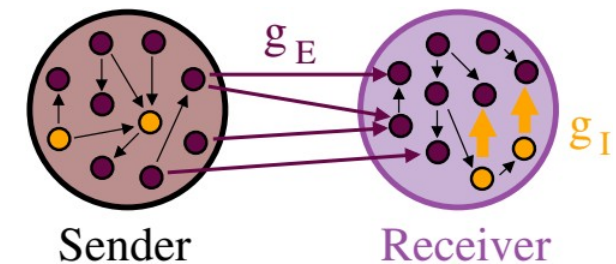
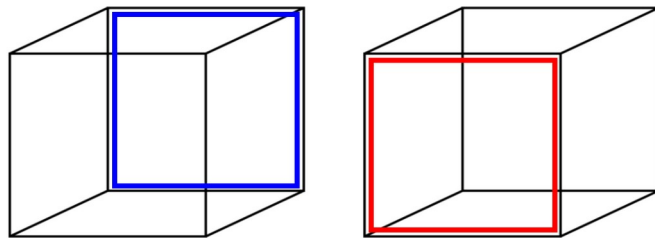
Phase bistability and bistable perception

We propose a neuronal model that presents phase bistability between positive and negative phases

PHYSICAL REVIEW E **102**, 032412 (2020)

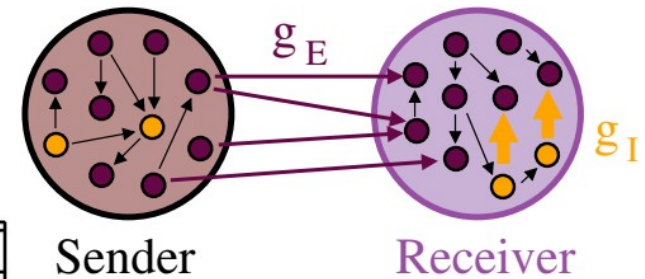
Phase bistability between anticipated and delayed synchronization in neuronal populations

Júlio Nunes Machado  and Fernanda Selingardi Matias 
Instituto de Física, Universidade Federal de Alagoas, Maceió, Alagoas 57072-970, Brazil

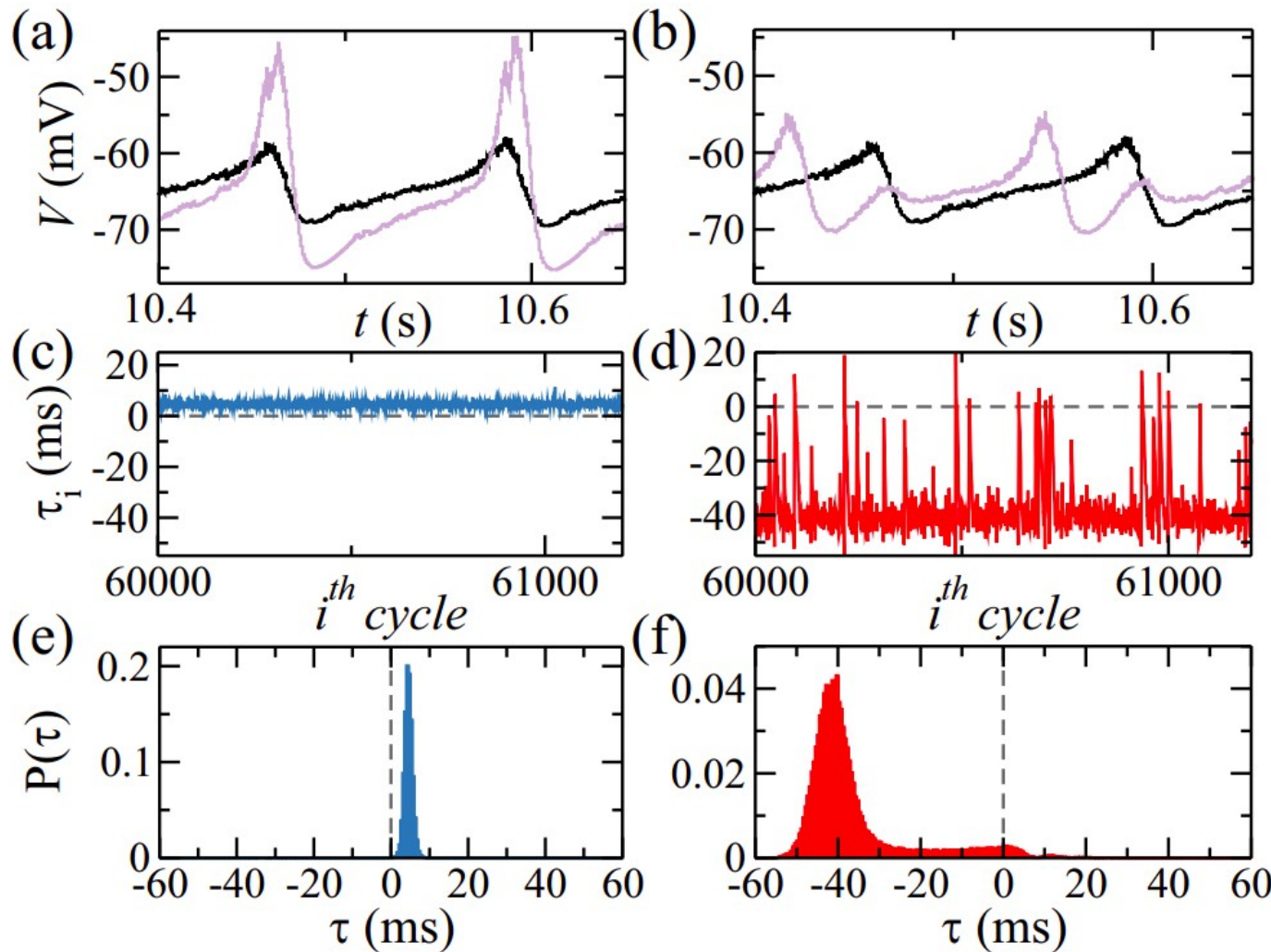


MEG results suggests that phase relations encode bistable perception (Kosen et al. 2014)

Anticipated Synchronization (AS)



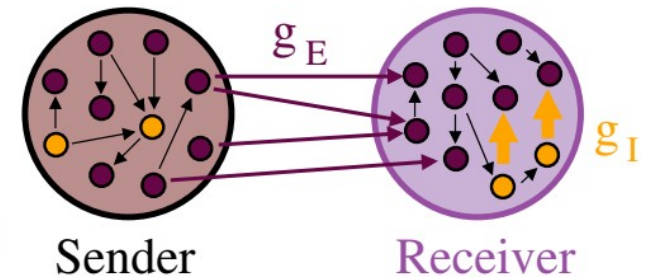
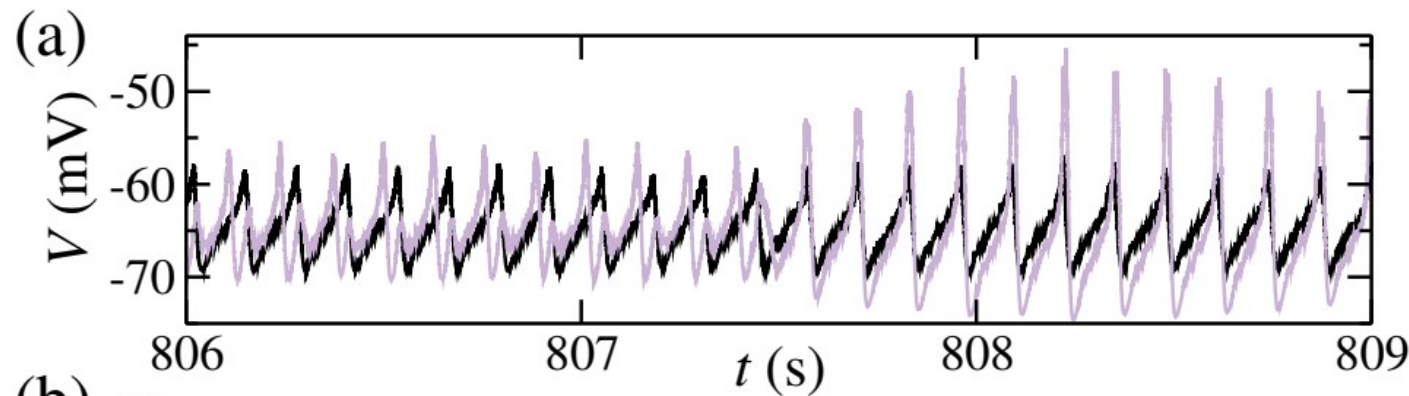
$$\tau_i \equiv t_i^R - t_i^S$$



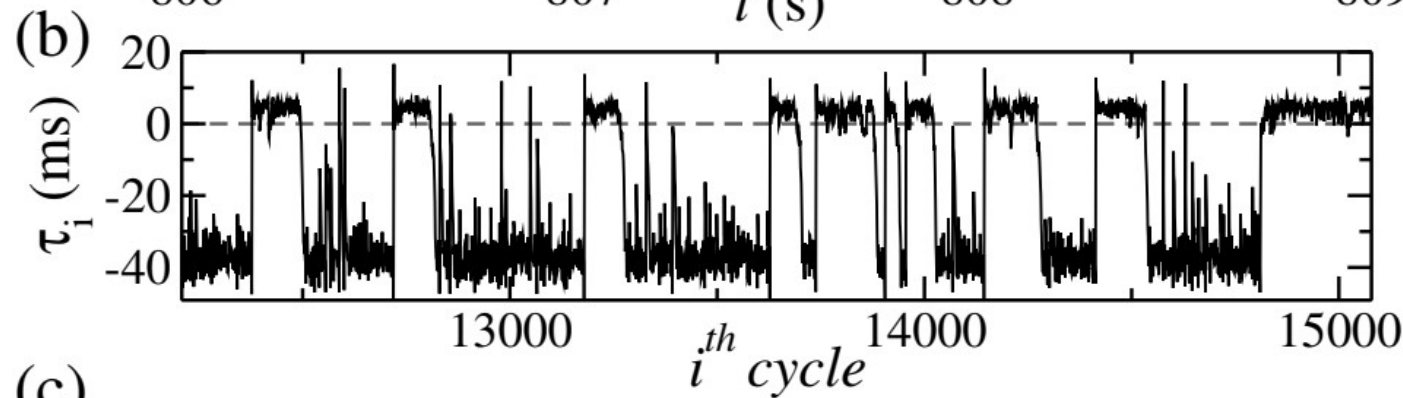
(DS)

(AS)

Phase bistability

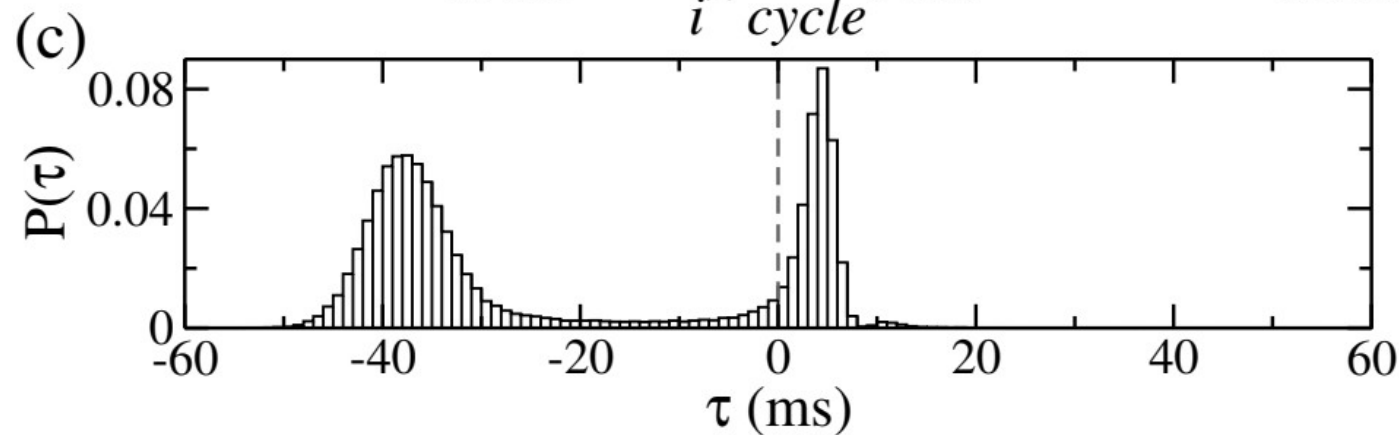


$$\tau_i \equiv t_i^R - t_i^S$$



(DS)

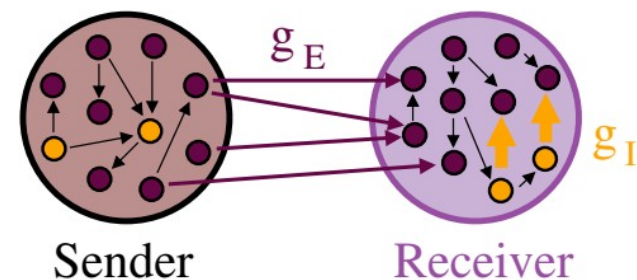
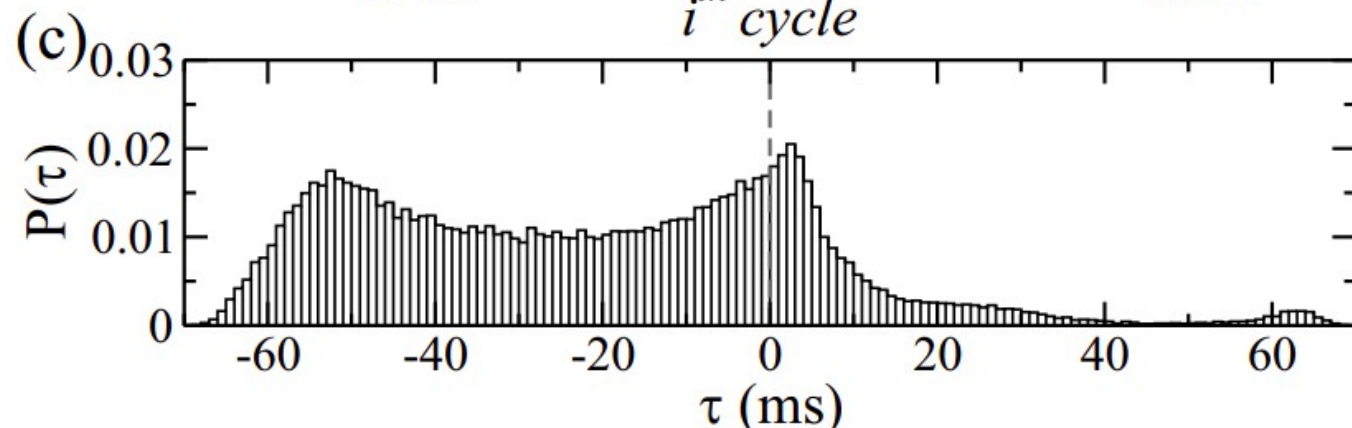
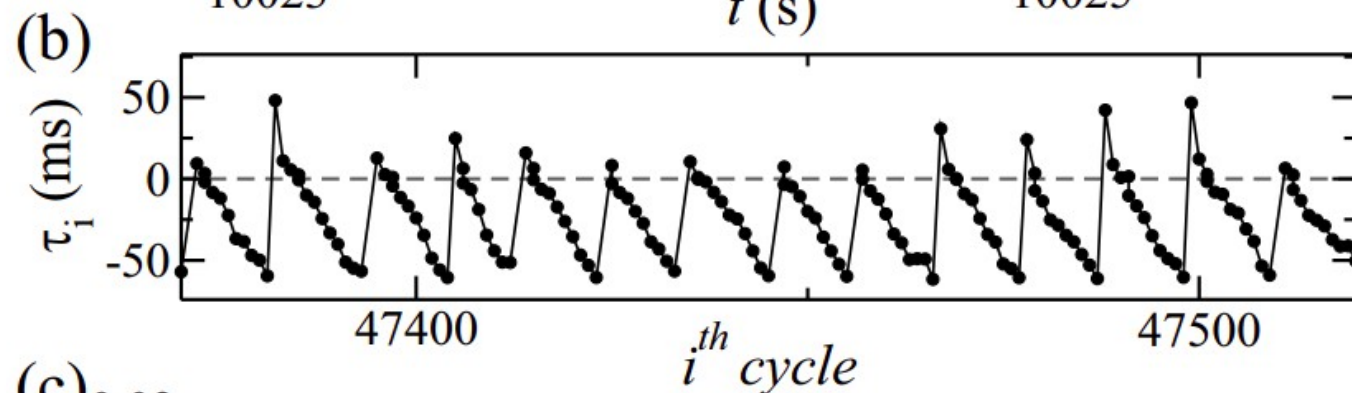
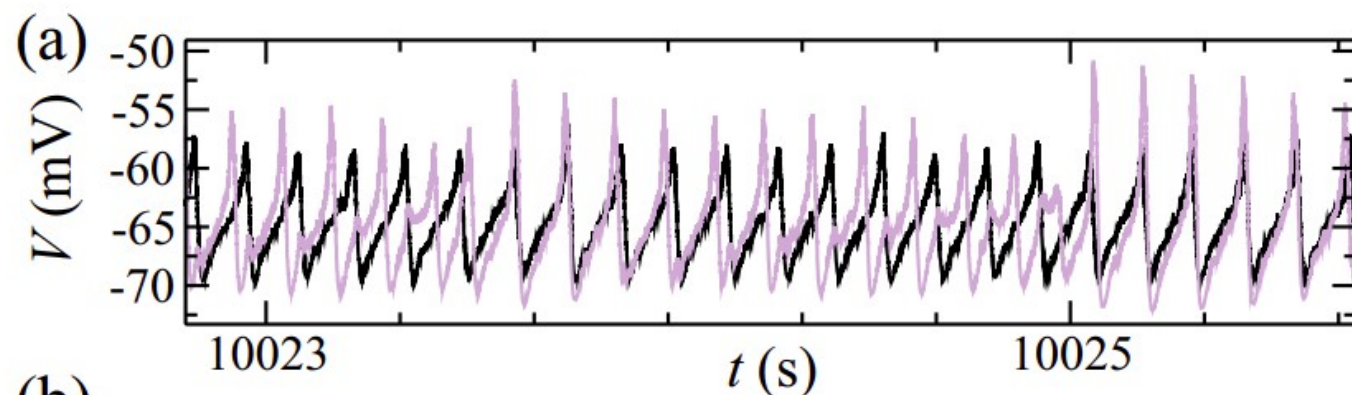
(AS)



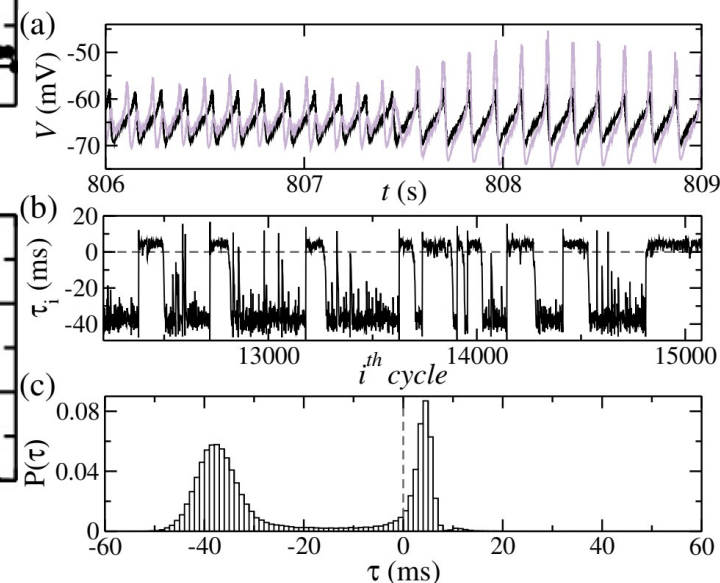
(AS)

(DS)

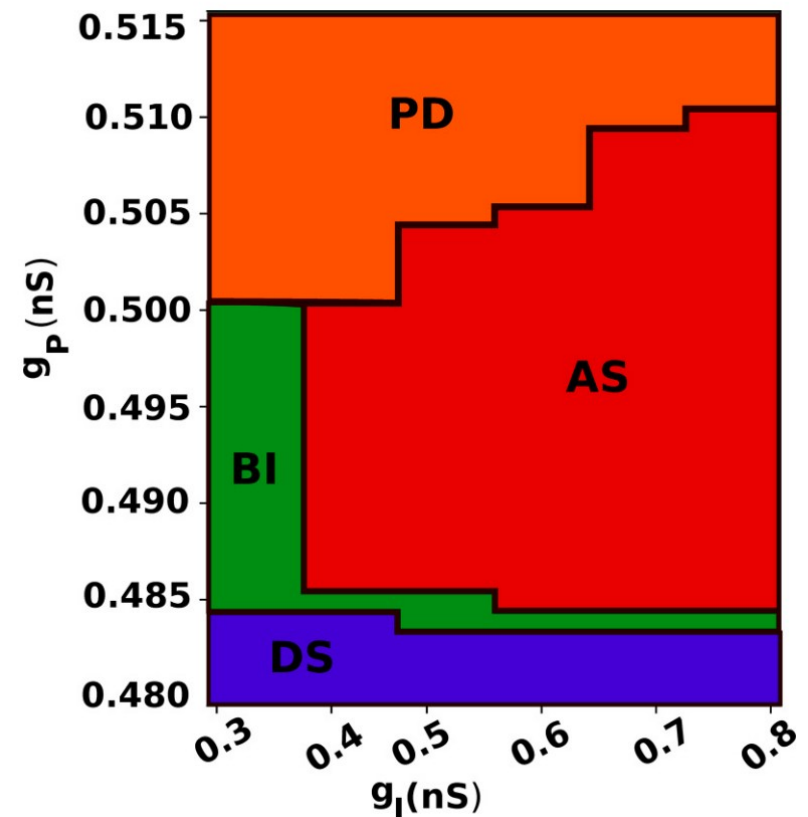
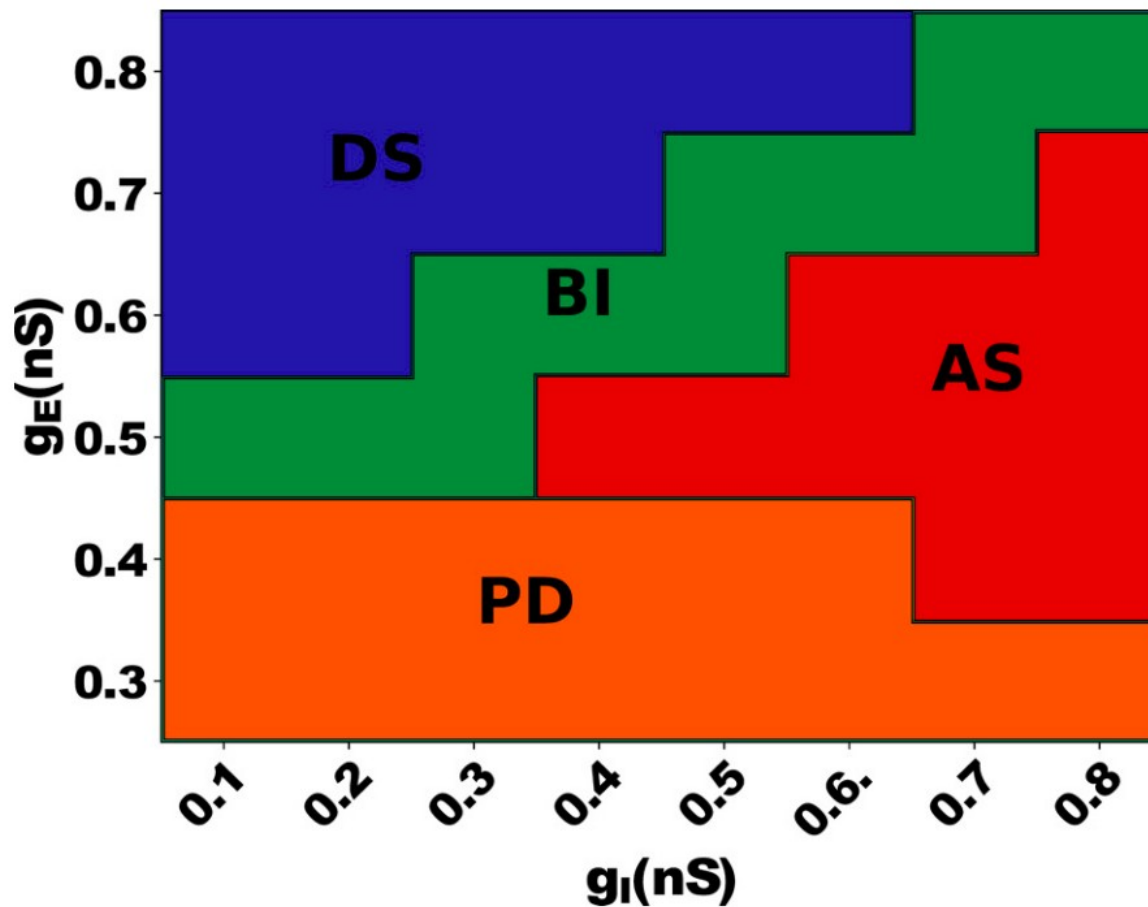
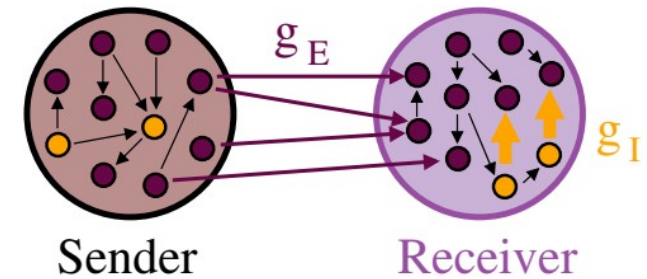
Phase-Drift



$$\tau_i \equiv t_i^R - t_i^S$$



A combined effect of excitation, inhibition and noise



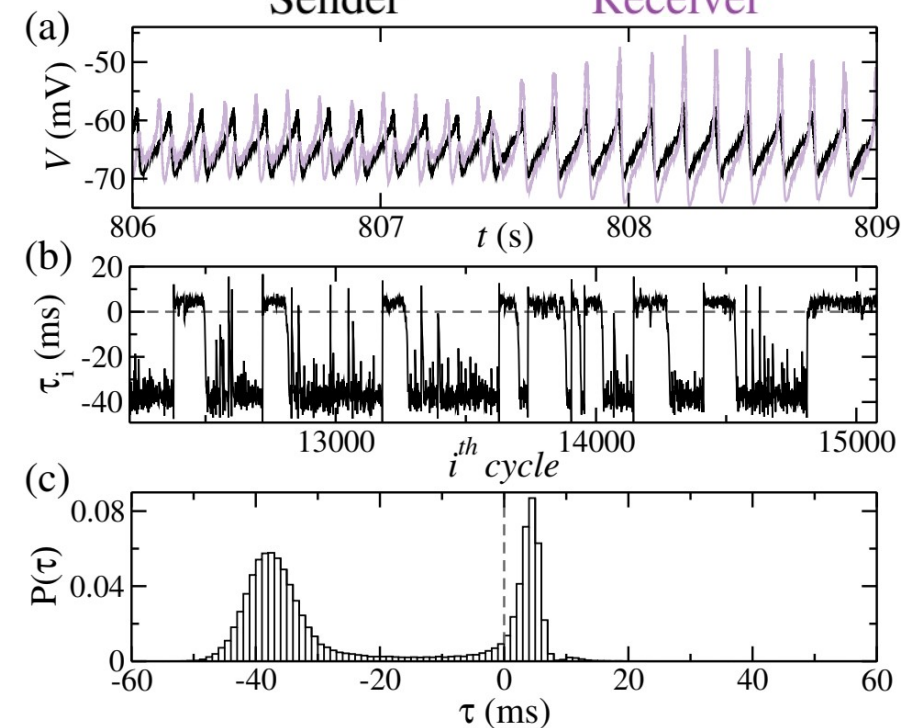
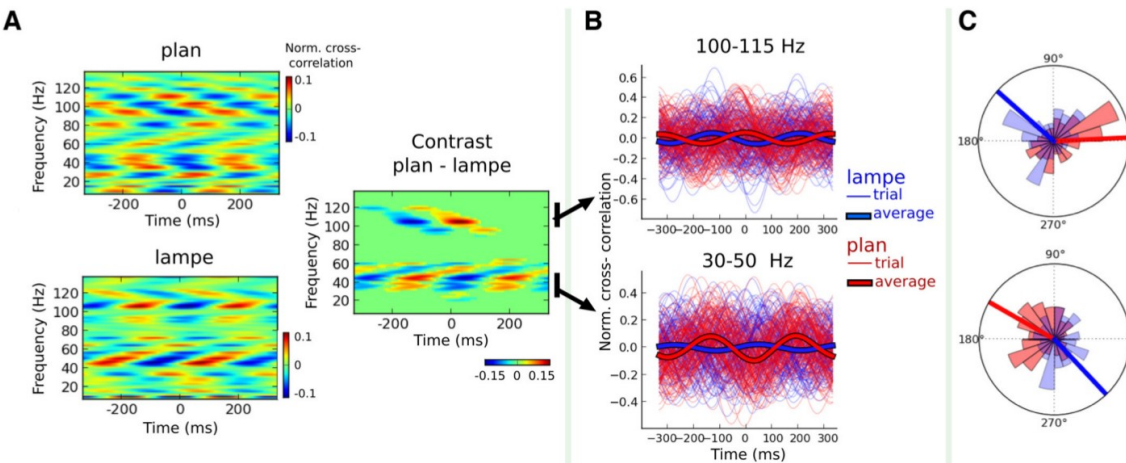
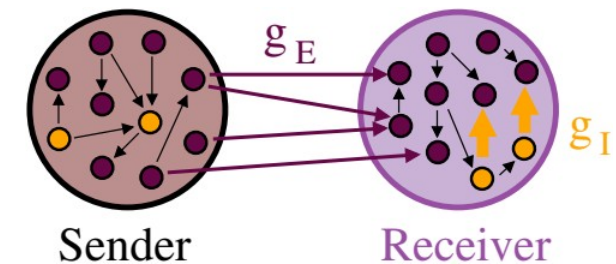
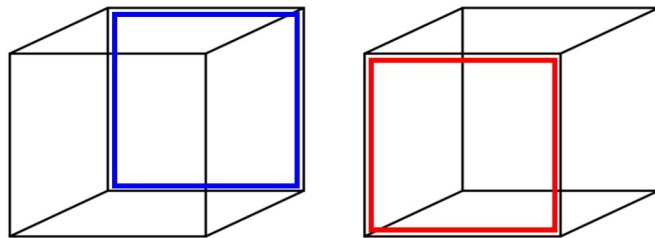
Phase bistability and bistable perception

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Phase bistability between anticipated and delayed synchronization in neuronal populations

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MEG results suggests that phase relations encode bistable perception (Kosen et al. 2014)

Phase bistability and bistable perception

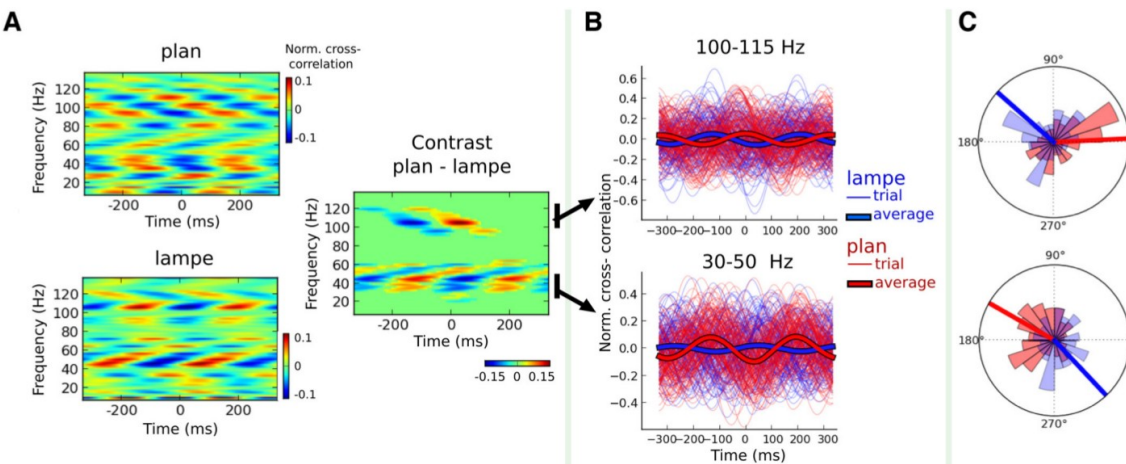
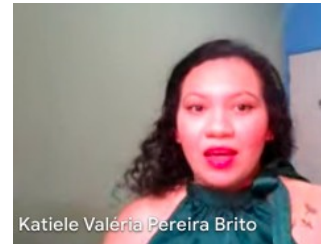
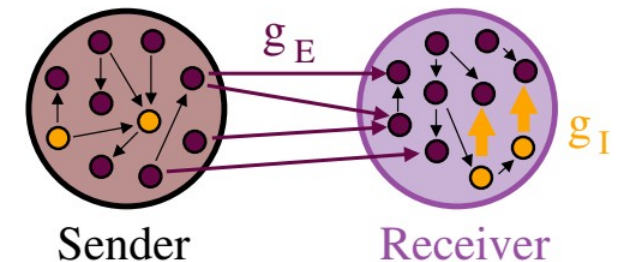
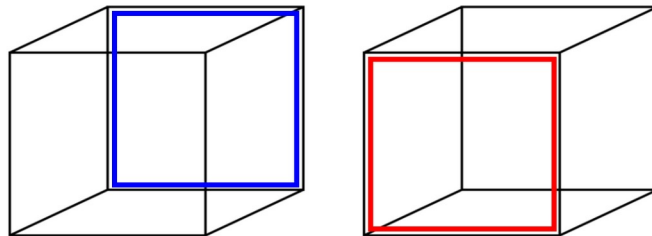
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Izhikevich Neuron Model
(neuronal variability)

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + \sum I_x,$$

$$\frac{du}{dt} = a(bV - u).$$

Synaptic current AMPA and GABAA

$$I_x = g_x r_x (v - E_x)$$

$$\tau_x \frac{dr_x}{dt} = -r_x + \sum_k \delta(t - t_k)$$

Each neuron receive a Poisson input

Neuronal variability & E/I balance:

Brito & Matias PRE (2021)

Thank you!



Mulheres nas Ciências Exatas,
Engenharia e Computação
Chamada CNPq-MCTIC No. 31/2018





NeuroLab

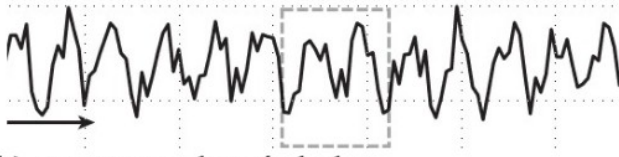


Mulheres nas Ciências Exatas, Engenharia e Computação
Chamada CNPq-MCTIC: No.31/2018

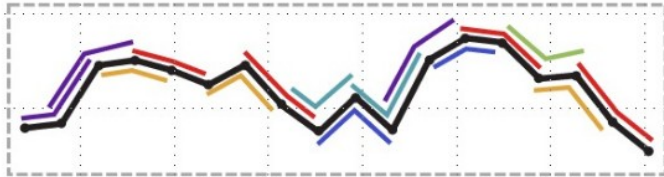
BRILHANTE MENTE



(a) Série temporal



(b) Contagem dos símbolos



(c) Histograma da ocorrência de cada símbolo

